

Signaling and Migrant Labor Market Integration: Experimental evidence from Colombia

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Abstract

This paper studies whether government-issued skill certification improves labor market integration when workers' skills are hard to verify. We evaluate a randomized offer of access to an occupation-specific certification program in Colombia, implemented among locals and migrants. The program certifies existing skills rather than providing training. Among locals, the offer increases employment by reducing inactivity and raises hours worked. Among migrants, it reduces skill downgrading without increasing wages or overall employment. The effects persist unevenly over time and are robust to attrition bounds and alternative outcome measures. The results show that one signal can affect different margins across groups.

Keywords: Signaling, Certifications, Skills, Migration, Colombia

JEL: J24, J61, J64, I26, O15, D82.

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1 Introduction

Employers often have limited information about workers' skills. Job seekers therefore rely on signals such as degrees, completed courses, grades, and institutional reputation (Spence, 1973). Information frictions may be especially relevant for migrants, whose education and work experience are often less familiar to employers in destination labor markets. Migrants may therefore face difficulties finding stable employment, work in jobs that do not match their qualifications (Dustmann et al., 2012; Lebow, 2023), or remain concentrated in informal employment (Delgado-Prieto, 2025). In some developing countries, labor market integration has remained incomplete despite regularization efforts (Bahar et al., 2021; Ibáñez et al., 2024).

This paper studies whether a credible skill signal improves labor market integration. We focus on skill certification, a policy that verifies existing competencies rather than providing training. We estimate the effects of an offer of access to an occupation-specific certification program for both locals and migrants. This design allows us to test whether the same signal affects workers with different ex ante signaling capacity along different labor market margins.¹

We study a free certification program implemented by the Colombian Ministry of Labor. The program certifies occupation-specific skills acquired through work experience or on-the-job training rather than formal education. Eligibility requires applicants to document prior work experience in the occupation linked to the certification. Among 2,160 eligible applicants, 855 randomly received an offer of access to the program, including 250 migrants. Treatment-assigned applicants were offered competency assessments and, if successful, certification. Control-group applicants were not assessed and did not receive certification during the evaluation period. Treatment and control applicants were surveyed at three, six, and twelve months after the intervention.

We estimate intention-to-treat effects separately for locals and migrants. Random assignment produced balanced treatment and control groups within both samples. Survey response rates are high, but treatment assignment increased follow-up response in some waves, especially among migrants. We report Lee bounds (Lee, 2009) to assess the sensitivity of the main estimates to differential attrition.

The effects differ across groups. Among locals, the offer of access to the certification program increases employment by 5.7 percentage points and reduces inactivity by 3.5 percentage points. It also raises weekly hours worked by 2.0 hours and full-time work by 3.7 percentage points, but does not increase formal employment or wages among those already employed. For migrants, the main effect is an 8.1 percentage point reduc-

¹The study was conducted under a pre-analysis plan accepted as part of the registered-report process, which specified the randomized offer of access to the certification program, separate analyses for locals and migrants, primary labor-market outcomes, and analyses of channels.

tion in skill downgrading, relative to a control-group mean of 25 percent, and is robust to alternative definitions of the outcome. This improvement in occupational match is not accompanied by higher wages. Dynamic estimates show that the local employment effect emerges quickly and remains positive after one year, while the migrant effects are concentrated in reduced skill downgrading and a short-run increase in formal employment. Although locals and migrants differ along baseline characteristics, the cross-group differences in employment and skill-downgrading effects are statistically significant, and reweighting the migrant sample to match locals on observables leaves the heterogeneous pattern unchanged. These results suggest that the same certification can operate on different margins: labor market entry for locals and match quality for migrants.

The evidence on additional outcomes is consistent with these different margins of adjustment. For locals, the program reduces unemployment duration by 0.23 months, without detectable changes in reservation wages or search intensity. This pattern is consistent with certification reducing screening frictions and facilitating labor market entry. For migrants, the evidence is more suggestive. The pooled reservation-wage effect is small, but reservation wages rise among unemployed migrants. This pattern is consistent with greater selectivity in job search, which may help explain why certification reduces skill downgrading without increasing overall employment. We find little evidence that changes in education or training enrollment explain the effects for either group over the period we study.

Compliance was high but not perfect. A small share of participants randomized into treatment did not receive a certificate because they did not successfully complete the evaluation process or because they withdrew. We therefore examine whether the intention-to-treat estimates are sensitive to incomplete compliance by instrumenting receipt of certification with random assignment to treatment. The instrumental variable estimates closely mirror the intention-to-treat estimates, as expected given the high compliance rate. Stated willingness to participate in the program without subsidies is high in both treatment and control groups, suggesting that participants valued the program regardless of assignment.

This paper contributes to the literature on labor market signaling. This literature shows that credible information about workers' skills can affect hiring and labor market outcomes. Experimental studies find positive effects from certification or information disclosure among young job seekers in Ethiopia ([Abebe et al., 2021](#)), during job interviews in Uganda ([Bassi and Nansamba, 2022](#)), among young job seekers in South Africa ([Carranza et al., 2022](#)), among young workers in New York ([Heller and Kessler, 2021](#)), and in online labor markets ([Pallais, 2014](#)). Related evidence shows that labor market signals such as grade point averages, college reputation, institutional quality,

and publicly available skills information can generate substantial returns (Landaud et al., 2024; MacLeod et al., 2017; Eble and Hu, 2022; Busso et al., 2025). We contribute by studying whether a common, occupation-specific skill certificate has different effects for locals and migrants. This allows us to test whether signaling has heterogeneous returns for groups with different ex ante signaling capacity. We find that the same signal affects different margins: it increases employment among locals but reduces skill downgrading among migrants.

The study also contributes to the literature on migrant labor market integration. This literature analyzes the effects of policies that reduce barriers to employment in destination labor markets, including job search assistance (Battisti et al., 2019), intensive coaching (Joona and Nekby, 2012), wage subsidies, training, and placement programs (Dahlberg et al., 2024; Foged and Van der Werf, 2023; Foged et al., 2024a), citizenship application subsidies (Hainmueller et al., 2018), mentorship programs (Jaschke et al., 2022), refugee-firm matching (Loiacono and Silva-Vargas, 2023), and regularization policies (Bahar et al., 2021; Ibáñez et al., 2024). These interventions can improve employment, earnings, legal status, or social integration, but evidence from Colombia suggests that regularization alone had limited effects on the labor market integration of Venezuelan migrants, especially among lower-skilled workers (Bahar et al., 2021; Ibáñez et al., 2024). We contribute by studying a policy that targets a different margin: migrants' ability to make existing skills legible to employers. Unlike training or intensive job search programs, skill certification does not aim to build new skills. It verifies existing competencies and provides a credential. Other forms of recognition, such as the validation of foreign degrees or professional credentials, can be costly and time-consuming. In their absence, migrants may retrain unnecessarily, generating productivity losses (Carranza and McKenzie, 2024). We find that an occupation-specific certificate reduces skill downgrading among migrants, suggesting that certification can improve labor market integration through better occupational matching even when standard employment margins do not move.

The rest of the paper is organized as follows. Section 2 describes the program setting and intervention. Section 3 presents the research design, data, outcomes, and empirical strategy. Section 4 assesses internal validity, including balance and attrition. Section 5 reports the main results and their dynamics. Section 6 presents robustness checks. Section 7 discusses certification receipt and heterogeneity. Section 8 examines channels and additional outcomes. Section 9 concludes.

2 Program Setting and Intervention

Colombia provides a useful setting to study whether certification can make existing skills more legible to employers. The country received a large inflow of Venezuelan migrants over a short period, many of whom obtained legal status and could participate in the labor market.² Despite their higher labor force participation, Venezuelan migrants face substantial occupational mismatch: across the region, 24 percent of highly educated migrants work in low- or medium-skilled jobs, compared to 17 percent of native-born workers (Kaplan et al., 2023). Credential non-recognition and employer uncertainty about migrant qualifications push many into informal or low-skilled jobs below their competency level (Bahar et al., 2021; Lebow, 2023). Colombia also has a long-standing system of skills certification, allowing us to study whether an occupation-specific domestic signal affects labor market outcomes.

2.1 Skill Certification Program

Since 2003, the Colombian government has operated a free skill certification program through the Colombian Vocational Training Agency (SENA).³ The program allows adults with prior work experience to validate existing occupational skills through publicly recognized certifications. It does not provide training. Instead, the certifying agency assesses whether applicants have already acquired specific competencies through work experience.⁴

The certifications can credibly signal workers' skills to Colombian employers for three reasons. First, the program certifies more than 200,000 workers each year throughout the country, making the credential widely used. Second, SENA is a recognized national institution that provides both certification and technical training, although these services are administered separately. Third, certification assessments are based on occupational standards developed with input from industry associations, giving employers a role in defining the competencies being certified. These features make the certificate a potentially informative signal of occupation-specific skills.

Eligibility requires a valid identification document, registration through the program website, acceptance of the legal terms, and at least six months of prior work experience in the occupation linked to the certification. The program offers assessments in 2,405 competencies across manufacturing, agriculture, retail, and service occupations. Applicants may be employed or unemployed, but they must select a specific compe-

²See Appendix A for details on the Venezuelan migration crisis, the regularization process, and migrants' labor market integration in Colombia and across the region

³The program is named, in Spanish, "Programa de Evaluación y Certificación de Competencias Laborales", which was regulated by Decree 933 of 2003.

⁴Appendix B.1 provides additional details about the skills certification program.

tency for certification and document prior practical experience in the corresponding occupation.

Candidates are evaluated through a practical assessment and a knowledge test. In the practical assessment, a qualified assessor from the certifying agency evaluates whether the applicant can perform occupation-specific tasks and whether they meet the competency standards, assigning a pass/fail grade. Applicants who pass this assessment then complete a multiple-choice knowledge test administered virtually, with scores ranging from 0 to 100. Successful candidates receive a certificate that reports their assessed level: basic, intermediate, or advanced.⁵ Candidates who do not pass are eligible for a six-hour remedial session and may subsequently retake the assessment. Historically, fewer than 10 percent of participants fail, consistent with the requirement that applicants document prior practical experience in the occupation before certification.

2.2 “Skills Pay Off” Program

Since 2021, *Saber Hacer Vale* (“Skills Pay Off”) has extended access to the certification program to vulnerable populations. Implementation begins with the selection of a geographic area and the competencies to be certified. The agency then determines the number of available slots, opens the call for applications, and verifies applicants’ documentation to determine eligibility.⁶

The program preserves the certification process described above but adds two components: information on publicly available job-search resources and financial assistance during participation. The information resources are unlikely to drive any treatment effects, as they are publicly available to both the treatment and control groups, participation is voluntary, and they constitute light-touch interventions ([Carranza et al., 2022](#); [McKenzie, 2017](#); [Kelley et al., 2024](#)). The financial assistance includes food and connectivity subsidies of approximately \$82 USD for all participants, with additional transportation, dependent-care, or remedial-session support for those who qualify. Total support can reach \$217 USD, equivalent to roughly 70 percent of the monthly minimum wage. Because benefits are paid upon completion, the payment structure creates incentives to start and complete the process. These resources are meant to compensate for lost income during the certification process, but we consider them

⁵The type of certificate awarded depends on the grade obtained in the evaluation process: those with scores below 30 fail the exam and do not receive the certificate; those with scores between 30 and 60 receive a basic certificate; those between 60 and 90 receive an intermediate certificate; and those above 90 receive an advanced certificate. The issued certificate explicitly states the level at which the individual was classified.

⁶Appendix B provides additional details about “Skills Pay Off”.

unlikely to affect labor supply decisions.⁷

The job-search resources are publicly available and involve limited interaction or support. The subsidies are tied to participation and completion and were intended to compensate participants for the opportunity cost of time during the certification process. In this sense, the program is best understood as a certification program that reduces the cost of participation for low-income individuals.

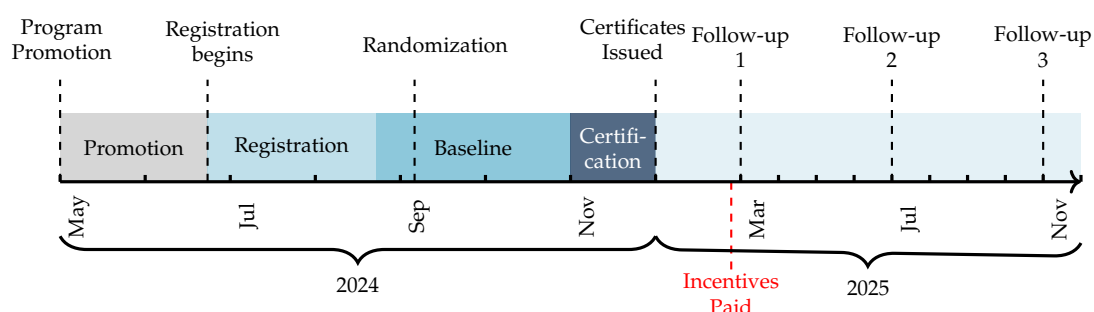
Eligibility for additional support requires meeting at least one vulnerability criterion, such as low income, caregiving responsibilities, age, ethnic-minority status, or being a migrant. At least 30 percent of available slots are reserved for migrants with regular status.

2.3 Intervention

The intervention evaluated in this paper is the 2024 cohort of “Skills Pay Off,” implemented in Colombia’s Pacific region.⁸ The cohort had 855 available slots across five competencies: health services, food preparation, food handling, design and tailoring, and recycling operations.

Figure 1 summarizes the implementation and evaluation timeline. The program was advertised in May and June 2024, and applications were open between June and August 2024. A total of 2,736 individuals registered through the online application form. Of these, 2,160 met the eligibility criteria, which primarily verified prior work experience in the occupation linked to the requested certification.⁹

Figure 1: Timeline of the Intervention



Applicants offered access to the program completed the standard certification process, which included a practical assessment administered by a qualified evaluator and a multiple-choice knowledge test from the certifying agency. Applicants who met

⁷Appendix B provides a detailed description of the publicly available job-search resources and the financial assistance.

⁸Specifically, in the departments of Valle del Cauca, Cauca, and Nariño.

⁹Submitted documents were reviewed between August 1st and September 3rd to determine eligibility for the program.

the competency standards received an official certificate indicating their assessed skill level. Applicants who did not initially meet the competency standards were offered a remedial session and could retake the assessment.¹⁰ Certificates were issued between November and December 2024.

Table 1 reports certification outcomes among applicants assigned to treatment. Compliance was high but incomplete: 89 percent of treatment-assigned applicants received a certificate. Certification levels were 15 percent basic, 23 percent intermediate, and 51 percent advanced. The remaining 11 percent did not receive a certificate: four applicants were assessed but did not meet the certification criteria, while 89 withdrew before completing the assessment process.

Table 1: Certification Assessment Outcomes and Treatment Assignment by Migration Status

Treatment Status	Local Sample		Migrant Sample	
	(n)	(%)	(n)	(%)
Advanced	333	19.9	104	21.5
Intermediate	114	6.8	84	17.4
Basic	87	5.2	40	8.3
Not competent	3	0.2	1	0.2
Withdrawn	68	4.1	21	4.3
Control	1,071	63.9	234	48.3

Notes: The table reports the distribution of participants by migration status. Advanced, Intermediate, Basic, and Not Competent refer to the proficiency level at which each treated participant was classified after completing the evaluation and certification process. Withdrawn denotes individuals assigned to treatment who did not complete the process. Control denotes individuals randomly assigned to the control group.

3 Research design

3.1 Randomization

Before 2024, access to the program was allocated on a first-come, first-served basis. In the 2024 cohort studied in this paper, excess demand led the Ministry of Labor to allocate access by lottery. On September 6th, 2024, the 855 available slots were randomly allocated among the 2,160 eligible applicants.

Randomization was conducted at the individual level, stratified by type of certification (i.e., competency) and whether applicants were migrants, with 30 percent of slots reserved for migrants. Individuals assigned to treatment were offered access to the certification program as implemented. Individuals assigned to control were

¹⁰Participants who completed the remedial session and subsequently passed the assessment received the certification and the corresponding financial incentives.

informed that they would be offered access to the certification program in a later round, approximately 12 months after the intervention began.¹¹

3.2 Data Collection

Participants assigned to treatment and control were surveyed before and after the intervention. Baseline data come from two sources. The registration form collected demographic, socioeconomic, educational, and employment characteristics at the time of application. Before the intervention, participants were also contacted by phone in September 2024 for a complementary baseline survey, which collected additional information on labor market outcomes, mental health, and poverty.

We then conducted three follow-up surveys, at three, six, and twelve months after the program ended. In each round, participants were recontacted by phone and invited to complete a survey comparable in scope to the baseline. The first follow-up took place between February and March 2025; the timing of all survey rounds is shown in Figure 1.¹²

3.3 Conceptual Framework and Analysis Plan

The analysis is guided by a simple conceptual framework and anchored in a pre-analysis plan. Employers may have imperfect information about workers' skills, especially when those skills were acquired through work experience rather than formal education. In this setting, an occupation-specific certificate can affect labor market outcomes if employers view it as credible information about workers' skills. Because the certification program assesses existing competencies rather than providing training, short-run effects are most naturally interpreted as changes in workers' ability to signal skills, employers' ability to screen applicants, or workers' job-search behavior.

The framework implies that the same signal may affect locals and migrants along different margins. For locals, whose education and work histories are more familiar to Colombian employers, certification may reduce screening frictions and increase employment or hours worked. For migrants, the value of the certificate may be greater for match quality than for employment entry. Migrants often hold foreign credentials and prior work experience that are less legible to local employers. A domestic certificate may therefore help them move into occupations that better match their qualifications, even if it does not increase overall employment.

¹¹The randomized allocation had to satisfy the Ministry of Labor's budget constraint.

¹²The pre-analysis plan anticipated a nine-month follow-up survey. This round was not collected because of budgetary and implementation-timing constraints. Although the first follow-up survey was fielded before the pre-analysis plan was accepted, the research team did not receive any follow-up data until after acceptance.

This logic shaped the analysis of outcomes and channels. The primary labor-market outcomes are employment, wages, and hours worked. We also study related employment margins, including inactivity, formal employment, and full-time work. For migrants, skill downgrading is central because it captures whether workers are employed in occupations below the skill level implied by their education. This outcome is directly tied to the hypothesis that certification can improve the allocation of migrant skills in the host labor market.

The framework also motivates the analysis of channels. Certification may affect outcomes by increasing labor demand, changing workers' reservation wages and job-search behavior, or inducing additional education, training, or credential acquisition. We therefore examine reservation wages, unemployment duration, search intensity, and subsequent enrollment in education, certification, or training. These outcomes do not identify employer beliefs directly, but they help distinguish whether the results are more consistent with reduced screening frictions, changes in job-search selectivity, or subsequent education and training responses.

3.4 Outcome Measures

The labor-market outcomes capture employment status and job quality. We measure whether respondents are employed, unemployed, or economically inactive, where inactivity is defined as not working and not actively seeking work. We also measure monthly wages, weekly hours worked, full-time work, and formal employment, defined as employment in which the worker contributes to the pension system. All these outcomes are measured directly in the surveys.

We also define *skill downgrading*, a central outcome for the migrant-integration hypothesis. This indicator equals one for individuals with post-secondary education who are employed in occupations requiring fewer qualifications than their educational level, and zero otherwise. To construct the measure, we use the Colombian Classification of Occupations (CUOC),¹³ which maps each occupation to a skill level based on the complexity and qualifications required to perform its tasks.

To study potential mechanisms, we collected two sets of additional outcomes. The first captures job-search behavior: whether the respondent received a job offer, the number of offers received, weekly hours devoted to job search, unemployment duration, and reservation wages. Reservation wages are elicited separately for employed and unemployed respondents as the minimum salary at which they would switch jobs or accept a job offer, respectively. The second set captures post-program education and

¹³The CUOC adapts the International Standard Classification of Occupations (ISCO-08) to the Colombian context.

training.

We also collect secondary outcomes on job satisfaction, well-being, and program valuation. Job satisfaction is measured by an indicator for reporting being satisfied or very satisfied with the current job. Well-being is measured by anxiety, using the GAD-7 scale, and food insecurity, using an indicator for having skipped a meal because of lack of economic resources. Program valuation is measured by respondents' willingness to pay for a similar skill certificate and willingness to participate in the program without subsidies.¹⁴

3.5 Empirical strategy

The primary estimand is the intention-to-treat effect (ITT) of being offered access to the certification program as implemented. The main specification pools the three post-treatment survey rounds:

$$y_{ijt} = \alpha_0 + \alpha_1 D_i + \mu_j + \mu_t + \varepsilon_{ijt}, \quad \text{for } M_i \in \{0, 1\}, \quad (1)$$

where y_{ijt} represents an outcome for individual i , enrolled in certification area j , in survey wave t ; D_i is an indicator equal to one if the individual was randomly assigned to receive an offer of access to the certification program; M_i is an indicator equal to one for migrants; and μ_j and μ_t denote competency and survey-wave fixed effects, respectively. The coefficient α_1 identifies the average intention-to-treat effect for the corresponding subgroup.

Because not all treatment-assigned applicants received a certificate, we also estimate IV effects of certification receipt. We instrument receipt of certification with random assignment to treatment. The resulting 2SLS estimand is a local average treatment effect (LATE) for compliers.

We also estimate time-specific effects to examine whether the impact of the offer of access to the certification program varies over time. This allows us to assess timing and persistence. The dynamic specification takes the form:

$$y_{ijt} = \beta_0 + \beta_1 D_i + \sum_{t > T_0}^T \beta_{2t} (D_i \times T_t) + \mu_j + \mu_t + \varepsilon_{ijt}, \quad \text{for } M_i \in \{0, 1\}, \quad (2)$$

where all terms are defined as in Equation (1). The indicators T_t denote follow-up survey waves. The coefficients β_{2t} capture how the treatment effect differs across follow-up waves relative to the omitted survey wave.

¹⁴Appendix C defines the outcome measures used in the paper, maps them to the pre-analysis plan, documents planned outcomes not reported in the final analysis, and describes the construction of derived outcomes, including skill downgrading.

Equations (1) and (2) are estimated separately for locals and migrants. These subgroup estimates are internally valid because randomization was stratified by whether applicants were migrants. For completeness, we also report pooled interacted specifications and tests of equality between local and migrant effects. However, differences in estimated effects across groups should be interpreted with caution. Locals and migrants differ in pre-treatment characteristics, and these characteristics may moderate the returns to certification. As a result, cross-group differences in treatment effects may reflect migration background, observable or unobservable moderators, or some combination of these factors. The design does not separately identify these channels.

In all cases, standard errors are clustered at the individual level. Because we estimate effects on multiple outcomes, we adjust inference using sharpened q-values following [Anderson \(2008\)](#). We compute q-values within outcome families in each table and separately by estimation sample. Standard errors are reported in parentheses, while q-values provide the multiple-testing adjustment used to interpret significance across related outcomes.

4 Internal Validity

4.1 Balance

We assess randomization balance by comparing baseline characteristics between treatment and control applicants separately for locals and migrants. The tests condition on competency fixed effects, reflecting the stratified randomization design. For descriptive purposes, we also compare locals and migrants.¹⁵

Treatment and control applicants are balanced across observable baseline characteristics within both samples. Only one test rejects equality at the five-percent level, a pattern consistent with chance given the number of characteristics tested. This supports treatment-control comparisons within the local and migrant samples.

Locals and migrants differ along several baseline dimensions. Migrants are younger, more educated, more likely to be employed, and more likely to experience skill downgrading at baseline. These differences do not threaten identification within each subgroup, but they imply that cross-group differences in treatment effects cannot be attributed mechanically to migrant background alone. They also motivate the paper's focus on skill downgrading as a central margin of migrant labor market integration. Migrants are more educated—18 percent hold university or postgraduate degrees compared to 8 percent of locals—yet report similar wages. Among migrants with professional education, 12 percent work in occupations requiring fewer skills than their

¹⁵Baseline balance tests are reported in Appendix [D](#).

qualifications, compared to 4 percent of locals.¹⁶

4.2 Survey Attrition

Attrition can be an important threat to internal validity after random assignment. Recontacting applicants after registration is difficult, and treatment assignment may affect survey response because treated applicants remain connected to the program while control applicants do not. To reduce non-response, all participants were offered monetary incentives for completing follow-up surveys, independent of treatment status.¹⁷ These incentives were designed to preserve engagement in both the treatment and control groups.

Overall response rates are high. About 90 percent of participants completed the complementary baseline, and more than 82 percent completed the twelve-month follow-up.¹⁸ However, response rates differ by treatment status in some waves. Differential attrition is moderate and concentrated among migrants: treatment-assigned migrants are more likely to respond at the three- and six-month follow-ups.

To assess the sensitivity of the estimates to differential attrition, we report bounds following [Lee \(2009\)](#). The approach trims the outcome distribution of the group with the higher response rate in proportion to the response-rate gap between treatment and control. It relies on a monotonicity assumption: treatment assignment weakly increases the probability of survey response. This assumption is plausible in our setting because treatment-assigned applicants remained connected to the program, while control applicants had no comparable program-related reason to respond. As a robustness check, we report the resulting bounds and discuss their implications for the main estimates.

¹⁶This pattern is less pronounced for the broader skill-downgrading measure that includes individuals with technical education, which we attribute to limited occupational detail at baseline.

¹⁷Participants received \$7 USD for each completed survey and could receive up to \$30 USD for completing all follow-up rounds. Participants who missed a round could still participate in later rounds and remain eligible for the full incentive if they completed the required survey modules.

¹⁸Response rates by treatment status and nationality, along with tests of differential attrition by migrant background and wave, are reported in Appendix E. Differences are statistically significant at the 1% level in the three-month follow-up wave, at the 5% level in the six-month wave, and at the 10% level in the twelve-month wave.

5 Results

5.1 Main Results

Table 2 reports pooled intention-to-treat estimates from Equation (1). The estimates pool observations across the three follow-up rounds and are reported for the full sample (Panel A), locals (Panel B), and migrants (Panel C). All specifications include competency and survey-wave fixed effects.

In the full sample, access to the certification program increases employment by 3.6 percentage points, a 7 percent increase relative to the control mean. The effect on the zero-imputed log wage is positive, consistent with the employment response. Conditional estimates restricted to employed workers show no corresponding wage gain, indicating that the earnings effect is driven by the extensive margin rather than by higher wages among those employed (-0.020 , $p\text{-value}=0.650$). We find no statistically significant effects on formal employment, weekly hours worked, full-time work, or skill downgrading in the full sample.

Among locals, access to the certification program increases employment by 5.7 percentage points and reduces inactivity by 3.5 percentage points, suggesting that part of the employment gain reflects transitions out of inactivity. Certifications also increase the zero-imputed log wage, weekly hours worked by 2.0 hours, and the probability of full-time work by 3.7 percentage points. The hours effect represents a 15.7 percent increase relative to the control mean reported in Table 2.¹⁹

The pattern of results differs for migrants.²⁰ Most estimates in Panel C are imprecisely estimated and not statistically different from zero. The main exception is skill downgrading: access to the certification program reduces skill downgrading by 8.1 percentage points, relative to a control-group mean of 25 percent. This corresponds to a reduction of roughly one-third relative to the control mean.²¹ Although this estimate does not survive the multiple-hypothesis adjustment, reflecting both the smaller migrant subsample and the broader family of tests, it is persistent across follow-up waves (Section 5.2), and robust to differential attrition (Section 6), supporting its substantive interpretation.

¹⁹The employment and log-wage effects for the full sample, as well as the employment, inactivity, log-wage, weekly-hours, and full-time effects for locals, retain significance at conventional levels after the multiple-testing adjustment (q-values below 0.05).

²⁰Appendix F reports tests of equality between the local and migrant estimates. We reject equality at the 5 percent level for employment and skill downgrading, and at the 10 percent level for the zero-imputed log wage and weekly hours worked. These results should be interpreted with caution, as cross-group differences may reflect moderating effects of baseline characteristics rather than migrant status per se.

²¹Estimates restricted to employed migrants are also negative and larger in magnitude (coefficient = -0.10 , $p\text{-value} = 0.043$), suggesting that the reduction in skill downgrading is not driven only by changes in employment composition.

Table 2: Effects of Access to Skill Certification on Labor Market Outcomes

Intention To Treat estimated using Equation (1)

Outcomes Replaced by Zero if Non-Employed							
			Log	Formal	Weekly		Skill
	Employed	Inactive	Wage	emp.	working	Full time	Downgrade
	(1)	(2)	(3)	(4)	hours	(7+ hrs)	(7)
Panel A: Full Sample							
Treatment	0.036**	-0.024	0.490**	-0.001	1.223	0.024	-0.016
	(0.018)	(0.015)	(0.245)	(0.013)	(0.784)	(0.016)	(0.015)
Q-value	0.011	0.013	0.011	0.267	0.011	0.011	0.266
Mean Control	0.530	0.240	6.734	0.128	14.75	0.256	0.156
Observations	5647	5647	5634	5647	5647	5647	5647
Panel B: Locals							
Treatment	0.057***	-0.035*	0.741***	-0.008	2.016**	0.037**	0.004
	(0.021)	(0.018)	(0.283)	(0.015)	(0.857)	(0.017)	(0.016)
Q-value	0.032	0.046	0.032	0.258	0.033	0.042	0.293
Mean Control	0.494	0.264	6.199	0.136	12.88	0.221	0.136
Observations	4404	4404	4392	4404	4404	4404	4404
Panel C: Migrants							
Treatment	-0.033	0.011	-0.339	0.023	-1.485	-0.020	-0.081**
	(0.035)	(0.024)	(0.485)	(0.025)	(1.828)	(0.037)	(0.035)
Q-value	1.000	1.000	1.000	1.000	1.000	1.000	0.192
Mean Control	0.705	0.127	9.323	0.0903	23.79	0.424	0.250
Observations	1243	1243	1242	1243	1243	1243	1243
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a skill certification offer, regardless of whether certification was completed (intent-to-treat). The table reports pooled ITT estimates from Equation 1, stacking three follow-up rounds at 3, 6, and 12 months. Panel A presents results for the full sample; Panels B and C report estimates separately for locals and migrants. Outcomes in columns (3)–(7) are set to zero when the individual is not employed; monthly earnings are set to zero before applying the log transformation $\log(\text{wage} + 1)$. All specifications include competence and wave fixed effects; full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Sharpened q -values are reported for Panels A, B and C and are computed within each outcome family following [Anderson \(2008\)](#). Mean Control reports the control-group mean in the corresponding estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Taken together, the results suggest different margins of adjustment for locals and migrants. For locals, access to the certification program increases employment and reduces inactivity, consistent with labor market entry. For migrants—who were more likely to be employed at baseline—the main effect is a reduction in skill downgrading, consistent with improved occupational matching rather than entry into employment. This pattern is consistent with the hypothesis that workers with weaker pre-existing signals may benefit from certification through improvements in match quality. However, the design does not determine whether the difference between locals and migrants reflects migrant status itself, observable or unobservable moderators, or compositional differences between the two samples.

5.2 Dynamic Effects

The pooled estimates may mask differences in timing and persistence. Effects may emerge quickly if access to the certification program improves job matching

during search, or more gradually if employers learn about certified workers over time. They may also attenuate if the informational value of certification declines as employers acquire direct information about workers' productivity through labor market interactions (Farber and Gibbons, 1996). Figure 2 reports intention-to-treat estimates from Equation (2) at three, six, and twelve months after the intervention, separately for locals and migrants.

For locals, the employment effect is positive in all three follow-up waves and remains statistically significant one year after the intervention. The reduction in inactivity is concentrated in the short run and fades by six months. The zero-imputed log wage follows a similar pattern to employment, consistent with earnings gains operating through the extensive margin rather than through wage increases among employed workers. Weekly hours worked also rise initially, but the effect declines from 3.5 additional hours at three months to 0.8 hours at twelve months. Overall, the dynamic pattern suggests that access to the certification program primarily accelerates labor market entry for locals, with the largest gains concentrated in the short run and persistent effects on employment after one year.

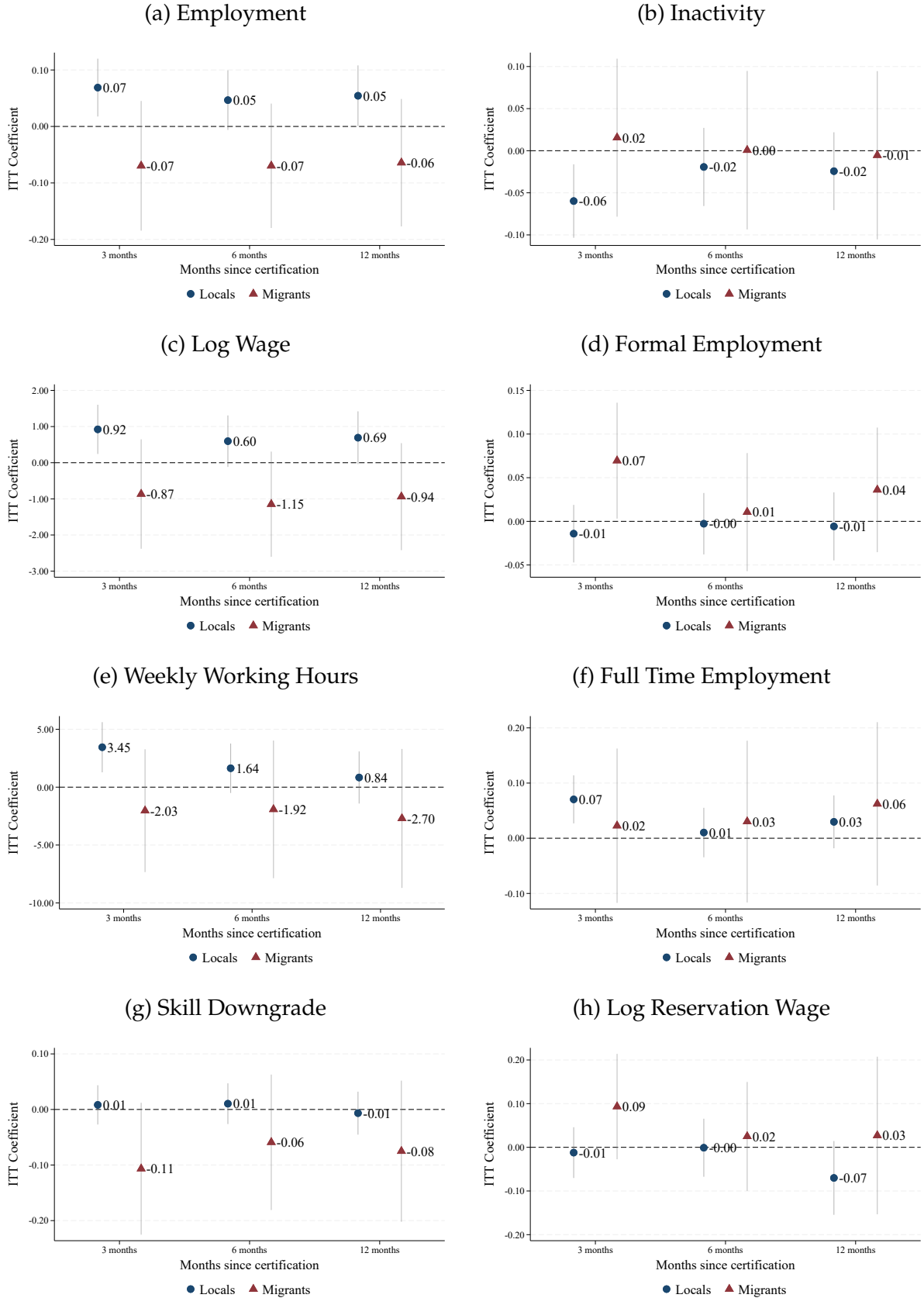
For migrants, the dynamic pattern differs from that for locals. The effects are concentrated in skill downgrading and, to a lesser extent, formal employment. Skill downgrading falls by 11 percentage points at three months and remains negative at six and twelve months. The estimates are imprecise in later waves, but the pattern is consistent with persistent reduction in downgrading.²² Formal employment increases by 7 percentage points at three months, relative to a control mean of 6.3 percent, but the effect is smaller at six and twelve months. We find no statistically significant effects on the remaining outcomes. Hence, the estimates suggest that access to the certification program affects migrants mainly through match quality rather than employment entry.

The dynamic evidence reinforces the pooled results. For locals, access to the certification program increases employment quickly, with effects that partly persist after one year. For migrants, the effects are concentrated in skill downgrading and short-run formal employment, consistent with improved match quality rather than employment entry. These patterns are consistent with a signaling interpretation, in which a certificate is most valuable when workers search for jobs and employers screen applicants.

²²The skill-downgrading coefficients are statistically indistinguishable across follow-up waves (p -value = 0.158 for Follow-up 1 vs. Follow-up 2, p -value = 0.319 for Follow-up 1 vs. Follow-up 3, and p -value = 0.629 for Follow-up 2 vs. Follow-up 3).

Figure 2: Effects of Access to Skills Certification on Labor Market Outcomes

Dynamic Intention To Treat Effects estimated using Equation (2)



Notes: Coefficients from Equation (2). Each panel shows ITT estimates of the effect of assignment to certification at 3, 6, and 12 months after the intervention. Marker values show point estimates. Bars represent 95% confidence intervals. Standard errors are clustered at the individual level. All specifications include competence and wave fixed effects.

6 Robustness

6.1 Sensitivity to Differential Attrition

Section 4 documents high overall response rates but moderate differential attrition, concentrated among migrants in the three- and six-month follow-up surveys. If non-responding control-group members had systematically better outcomes than respondents, the ITT estimates could overstate treatment effects. To assess this possibility, we follow Lee (2009) and estimate bounds for the main treatment effects, separately for locals and migrants.

We estimate ITT effects and Lee bounds for the full sample, locals, and migrants across the three follow-up waves.²³ For locals, the bounds are tight, reflecting limited differential attrition in this subgroup. At three months, both bounds for employment are positive and statistically significant, with a lower bound of 0.060 and an upper bound of 0.082. The corresponding bounds for inactivity are negative and statistically significant. The bounds for the zero-imputed log wage, weekly hours worked, and full time employment are also bounded away from zero at three months, although the lower bounds lose significance at six and twelve months. This pattern is consistent with the attenuation documented in the dynamic estimates. Overall, the bounds indicate that selective non-response is unlikely to explain the main employment effects for locals.

For migrants, the bounds are wider, reflecting larger differential attrition in this subgroup. The skill-downgrading result remains robust to this adjustment. At three months, both bounds are negative and statistically significant, with a lower bound of -0.180 and an upper bound of -0.090 . At six months, the lower bound remains negative and significant (-0.108 , $p\text{-value} < 0.01$). At twelve months, both bounds are negative; the lower bound is significant (-0.139), while the upper bound loses significance (-0.063). These results indicate that differential attrition is unlikely to explain the reduction in skill downgrading among migrants. By contrast, the formal-employment effect is less robust: at three months, the upper bound remains positive, but the lower bound becomes negative in later waves, consistent with the attenuation documented in Section 5.2.

6.2 Robustness of the Skill-Downgrading Measure

We assess the robustness of the skill-downgrading results using an alternative measure based on the observed educational composition of occupations in Colombia. The main measure assigns occupations to standardized skill levels and defines downgrad-

²³Results are reported in Appendix G.1.

ing by comparing those skill requirements with workers' educational attainment. This classification may not fully capture how occupations are allocated in the labor market: workers with different levels of schooling may hold similar jobs, and occupations classified as low-skill may employ many workers with post-secondary education.

We construct an alternative measure based on the observed educational composition of occupations in Colombia, following [Leuven and Oosterbeek \(2011\)](#). This measure equals one for participants with post-secondary education who are employed in occupations in which the modal worker has secondary education or less, and zero otherwise. We compute the modal education level within each occupation using Colombia's *Gran Encuesta Integrada de Hogares* (GEIH), the country's main household labor-force survey.²⁴ We use the 2010–2014 waves, before the large Venezuelan inflow into Colombia began in 2015, so that modal educational attainment within occupations is measured prior to the migration shock.²⁵

The estimates using the alternative skill-downgrading measure are consistent with the main results.²⁶ For migrants, the offer of access to the certification program reduces skill downgrading by 6.1 percentage points, compared with 8.1 percentage points under the main measure. The estimate is statistically significant at the 10 percent level. This similarity suggests that the skill-downgrading result is not driven by the use of standardized skill levels rather than the empirical educational distribution of occupations in Colombia.

7 Certification Receipt and Heterogeneity

7.1 Certification Receipt and Program Valuation

Stated preferences suggest that the program was valued by participants, but not differentially affected by treatment assignment. Approximately 88.2 percent of control-group respondents would participate in the certification program without subsidies. This share is similar among locals and migrants. Treatment assignment has no statistically significant effect on willingness to participate without subsidies. Willingness to pay is also positive for most respondents: 63.7 percent of control-group respondents

²⁴To ensure comparability between the GEIH and our survey data, we map the more aggregated GEIH occupation codes (based on ISCO-68) into our CUOC-based classification (ISCO-08). Given the greater aggregation of the GEIH coding, multiple CUOC occupation codes map into a single GEIH code. We restrict the GEIH sample to employed individuals of working age (15–64), yielding 1,682,333 observations.

²⁵This modal approach has been used to characterize occupational downgrading among Venezuelan migrants in Colombia by [Lebow \(2023\)](#), who reports that 52 percent of migrants in Colombia work in occupations where the modal worker is less educated than they are.

²⁶Estimates are reported in Appendix [G.2](#).

report a positive willingness to pay, with an average midpoint of 19.10 USD. The estimated treatment effects on willingness to pay and on the WTP midpoint are small and statistically insignificant in the full sample and within both subgroups. These results indicate that stated demand for certification was high, but that the offer of access to the certification program did not measurably change stated valuation.²⁷

Perceived program impact was collected only for treated participants in the first follow-up and is therefore not used in the experimental analysis. Descriptively, however, 64.0 percent of treated respondents reported that the certificate helped them greatly or moderately in finding a job.

Although stated willingness to participate without subsidies was high, program completion was not automatic. Of the individuals assigned to treatment, 89 percent obtained certification; the remaining 11 percent either withdrew before assessment or did not meet the certification criteria (see Table 1). We therefore examine whether the ITT estimates are sensitive to incomplete compliance, instrumenting receipt of certification with random assignment to treatment.

The IV estimates in Table 3 closely mirror the ITT estimates, as expected given the high compliance rate. Among locals, receipt of certification increases employment by 6.3 percentage points and raises the zero-imputed log wage; both estimates are significant at the 1 percent level. It also increases weekly hours worked by 2.2 hours and full-time work by 4.1 percentage points. As in the ITT estimates, receipt of certification has no statistically significant effect on formal employment or skill downgrading for locals. These results indicate that incomplete compliance does not explain the main local results.

For migrants, the IV estimate for skill downgrading is -0.088 and significant at the 5 percent level. This estimate is slightly larger in magnitude than the corresponding ITT estimate, as expected with high but incomplete compliance. Employment and inactivity effects remain small and statistically insignificant, and no other pooled estimate reaches conventional significance levels.

²⁷See Appendix Table H.1.

Table 3: Effects of Skill Certification on Labor Market Outcomes

Instrumental Variables Estimates

Outcomes Replaced by Zero if Non-Employed							
	Employed (1)	Inactive (2)	Log Wage (3)	Formal emp. (4)	Weekly working hours (5)	Full time (7+ hrs) (6)	Skill Downgrade (7)
<i>Panel A: Full Sample</i>							
Received certification	0.039** (0.020)	-0.026 (0.016)	0.538** (0.268)	-0.001 (0.014)	1.342 (0.859)	0.026 (0.017)	-0.018 (0.016)
Q-value	0.011	0.013	0.011	0.267	0.011	0.011	0.266
Mean Control	0.530	0.240	6.734	0.128	14.75	0.256	0.156
First stage F	7846	7846	7815	7846	7846	7846	7846
Observations	5647	5647	5634	5647	5647	5647	5647
<i>Panel B: Locals</i>							
Received certification	0.063*** (0.023)	-0.039** (0.020)	0.816*** (0.310)	-0.008 (0.017)	2.218** (0.941)	0.041** (0.019)	0.005 (0.018)
Q-value	0.032	0.046	0.032	0.258	0.033	0.042	0.293
Mean Control	0.494	0.264	6.199	0.136	12.88	0.221	0.136
First stage F	5530	5530	5502	5530	5530	5530	5530
Observations	4404	4404	4392	4404	4404	4404	4404
<i>Panel C: Migrants</i>							
Received certification	-0.036 (0.038)	0.012 (0.026)	-0.369 (0.527)	0.025 (0.027)	-1.615 (1.984)	-0.022 (0.040)	-0.088** (0.039)
Q-value	1.000	1.000	1.000	1.000	1.000	1.000	0.192
Mean Control	0.705	0.127	9.323	0.0903	23.79	0.424	0.250
First stage F	2658	2658	2655	2658	2658	2658	2658
Observations	1243	1243	1242	1243	1243	1243	1243
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Received certification (TOT) is an indicator equal to one for individuals who actually received a skill certification, instrumented by random assignment to the program. The table reports pooled TOT estimates obtained via IV/2SLS, using assignment to certification as the instrument for actual certification receipt. Panel A presents results for the full sample; Panels B and C report estimates separately for locals and migrants. Outcomes in columns (3)–(7) are replaced by zero for non-employed individuals. For non-employed individuals, monthly earnings are set to zero prior to the transformation $\log(\text{wage} + 1)$. All specifications include competence and wave fixed effects; full-sample regressions additionally include an indicator for migrant status. Standard errors (in parentheses) are clustered at the individual level. First stage F reports the Kleibergen-Paap rk Wald F-statistic for the excluded instrument, robust to clustering at the individual level. Sharpened q-values [Anderson \(2008\)](#) are computed from the corresponding ITT p-values separately for each sample over the family of main outcomes. Mean Control reports the control-group mean. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.2 Heterogeneity by Migrant Status

The IV estimates in Table 3 show that incomplete compliance does not explain the different margins of adjustment documented for locals and migrants. A separate interpretation issue is that these differences may partly reflect baseline characteristics rather than migrant status itself. Migrants are younger, more educated, and more likely to be employed at baseline. If these characteristics moderate the returns to certification, differences in estimated effects across groups may partly reflect differences in sample composition.

This concern is relevant because the two groups enter the experiment at different points in the labor market. Locals have lower baseline employment, leaving more scope for certification to affect labor market entry. Migrants, by contrast, are more likely to be

employed and more educated at baseline, but they are also more likely to experience skill downgrading. This pattern suggests that the relevant margin for migrants may be match quality rather than employment entry.

To assess the role of observable differences, we reweight the migrant sample using normalized propensity-score weights following [Busso et al. \(2014\)](#). The propensity score is estimated from a logistic regression of migrant status on demographics, education, baseline labor market outcomes, work experience, and competency of enrollment. The normalized weights rescale the inverse propensity scores so that they sum to one within each group, reducing the influence of extreme weights in regions of limited overlap.²⁸

After reweighting, baseline differences between locals and migrants are substantially reduced. Tests of equality between locals and the reweighted migrant sample yield *p-values* above 0.1 for all characteristics except the household head indicator, primary or less and tertiary education indicators, and the low-experience category.²⁹ The reweighted migrant sample is therefore more comparable to the local sample on observed baseline characteristics.

The reweighted ITT estimates are qualitatively similar to the baseline migrant estimates. The estimates continue to show a reduction in skill downgrading, with signs and magnitudes close to the baseline estimates, although they are not statistically significant. Employment and inactivity effects remain small and statistically insignificant.

Taken together, these results suggest that observable differences between locals and migrants are unlikely to account for the main heterogeneity documented in Section 5. The evidence remains consistent with the interpretation that an occupation-specific domestic signal may be especially valuable for migrants because foreign credentials and experience are less legible to local employers. However, this interpretation is indirect. Testing it directly would require data on employer perceptions or hiring behavior, which are not observed in this experiment.

8 Channels and Additional Outcomes

This section examines outcomes that help interpret the pattern of treatment effects. The results in Section 5 show that the offer of access to the certification program increased employment among locals and reduced skill downgrading among migrants. The outcomes below do not directly identify employer beliefs, but they indicate whether

²⁸Appendix [G.3](#) describes this exercise by comparing the reweighted sample of migrants to the sample of locals.

²⁹The Household head indicator remains imbalanced after reweighting (*p-value* = 0.09), as does the low-experience category (*p-value* = 0.01), the tertiary education (*p-value* = 0.06) and primary education indicators (*p-value* = 0.01). Excluding these variables from the propensity-score model does not materially affect the reweighted ITT estimates. See Appendix [G.3](#) for details on the reweighting exercise.

certification is associated with faster job finding, changes in search behavior, improved job matching, or subsequent education and training responses. Because the follow-up horizon is short, enrollment in education, training, or additional certification should be interpreted as a post-program response rather than as a mechanism that can fully explain the short-run labor-market effects.

8.1 Job Search, Education, and Training

Table 4 reports ITT estimates for job-search outcomes and post-program education and training. Job-search outcomes include the reservation wage, unemployment duration, and search intensity. Education and training outcomes include enrollment in additional education, other certification programs, and training courses. The pooled reservation wage combines the minimum salary at which unemployed respondents would accept a job and the minimum salary at which employed respondents would change jobs, controlling for follow-up employment status.³⁰ We discuss the estimates separately for locals and migrants.

For locals, the results suggest that employment gains are not driven by higher reservation wages or more intensive search. The estimated effect on the pooled log minimum acceptable wage is small and statistically insignificant (-0.027), with no evidence of meaningful differences by follow-up employment status. Search intensity is also unaffected, though this outcome is observed only for the unemployed and should be interpreted cautiously given the smaller sample. By contrast, the offer of access to the certification program reduces unemployment duration by 0.226 months. This pattern is consistent with certification reducing screening frictions and facilitating labor market entry. The offer also reduces enrollment in other certification programs by 4.3 percentage points and increases enrollment in training courses by 2.7 percentage points, with no statistically significant effect on enrollment in broader educational programs. These education and training responses do not appear to explain the short-run employment effects, but suggest some substitution away from additional certification and a modest increase in short training.

³⁰This approach follows [Abebe et al. \(2021\)](#) and [Carranza et al. \(2022\)](#).

Table 4: Effects of Access to Skill Certification on Reservation Wages, Job Search Behavior, Education, and Training

Intention-To-Treat estimated using Equation (1)

	Reservation Wage			Job Search Behavior		Education and Training		
	Pooled Res. wage (1)	Res. Wage Unemployed (2)	Res. Wage Employed (3)	Unemp. duration (months) (4)	Search intensity (hrs/week) (5)	Enrolled in additional education (6)	Enrolled in certification program (7)	Enrolled in training course (8)
<i>Panel A: Full Sample</i>								
Treatment	-0.016 (0.023)	-0.002 (0.035)	-0.042 (0.027)	-0.164** (0.083)	-0.133 (0.159)	-0.033** (0.016)	-0.045*** (0.004)	0.027** (0.011)
Q-value	1.000	1.000	0.436	0.004	0.349	0.011	0.001	0.011
Mean Control	14.25	14.11	14.38	1.894	1.875	0.267	0.0440	0.112
Observations	5509	2467	3042	5647	1588	5549	5549	5549
<i>Panel B: Locals</i>								
Treatment	-0.027 (0.027)	-0.032 (0.037)	-0.048 (0.033)	-0.226** (0.099)	-0.264 (0.170)	-0.030 (0.019)	-0.043*** (0.004)	0.027** (0.013)
Q-value	0.625	0.625	0.625	0.048	0.066	0.066	0.001	0.043
Mean Control	14.25	14.13	14.38	2.067	1.886	0.275	0.0417	0.111
Observations	4299	2092	2207	4404	1356	4333	4333	4333
<i>Panel C: Migrants</i>								
Treatment	0.028 (0.046)	0.177* (0.101)	-0.016 (0.045)	0.033 (0.144)	0.612 (0.443)	-0.037 (0.031)	-0.054*** (0.011)	0.028 (0.024)
Q-value	0.913	0.322	0.913	0.695	0.509	0.203	0.001	0.203
Mean Control	14.25	14	14.35	1.056	1.775	0.226	0.0548	0.113
Observations	1210	375	835	1243	232	1216	1216	1216
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Assigned to certification is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. The table reports pooled ITT estimates from Equation 1. Panel A presents results for the full sample. Panels B and C report estimates separately for locals and migrants. Column (1) reports estimates for the log minimum acceptable wage, defined as $\log(\text{reservation wage} + 1)$, pooling the minimum salary to accept a job (unemployed respondents) and the minimum salary to change jobs (employed respondents), controlling for employment status at follow-up. Column (2) restricts to individuals unemployed at follow-up and reports the minimum salary to accept a job offer. Column (3) restricts to individuals employed at follow-up and reports the minimum salary to change jobs. Unemployment duration is set to zero for employed respondents and is missing for inactive respondents. Search intensity was collected only in Follow-ups 2 and 3 and only among unemployed respondents. All specifications include competence and wave fixed effects. Full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Sharpened q-values [Anderson \(2008\)](#) are computed within each panel over three outcome families matching the column groups: reservation wages (Columns 1–3), job search behavior (Columns 4–5), and education and training (Columns 6–8). Mean Control reports the control-group mean in the corresponding estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

For migrants, the evidence on job-search behavior is more suggestive. The pooled effect on the log minimum acceptable wage is small and statistically insignificant. However, the decomposition by follow-up employment status shows a positive effect among unemployed migrants and no effect among employed migrants. This pattern is consistent with greater search selectivity among unemployed migrants, which could help explain why the offer of access to the certification program reduces skill downgrading without increasing overall employment. This interpretation is suggestive, however, because the decomposition conditions on post-treatment employment status. We find little evidence that education or training responses explain the migrant effects over the period we study. The offer reduces enrollment in other certification programs by 5.4 percentage points, has no statistically significant effect on broader educational enrollment, and increases training-course enrollment by 2.8 percentage points, although this

estimate is imprecise.

Additional labor-market outcomes provide a similar contrast across groups.³¹ Among locals, the offer of access to the certification program increases self-employment and employment in small firms, and it raises the probability of changing sector and changing jobs.³² These margins are consistent with the employment effects documented above: certification appears to facilitate labor market entry and broader job mobility among locals. For migrants, the estimates provide little evidence of broad changes in firm type, sectoral mobility, job changes, or job offers. This contrast reinforces the interpretation that the migrant effect is more targeted: certification does not generate broad employment reallocation, but it appears to improve the match between migrants' education and the occupations in which they work.

Taken together, the evidence on channels points to different margins of adjustment across groups. For locals, the offer of access to the certification program reduces unemployment duration and increases employment, without detectable changes in reservation wages or search intensity. This pattern is consistent with certification reducing screening frictions and facilitating labor market entry. For migrants, the evidence is more suggestive. The reservation-wage estimates are consistent with greater selectivity among unemployed migrants, while the main results show a reduction in skill downgrading. These patterns suggest that certification may have helped some migrants move into better-matched occupations rather than increasing overall employment. We find little evidence that additional education, training, or credential acquisition explains the effects for either group over the period we study.

8.2 Job Satisfaction and Well-Being

We find little evidence that the offer of access to the certification program affected job satisfaction or broader measures of well-being.³³ In the full sample, the estimates for job satisfaction, skipping a meal, and anxiety are small and statistically insignificant. Among locals, we find no corresponding improvement in job satisfaction, food insecurity or anxiety. Among migrants, the estimates are also statistically insignificant and do not point to systematic improvements in well-being. Overall, these results suggest that the program's detectable effects are concentrated in labor-market outcomes rather than broader short-run well-being.

³¹Appendix Table H.2 reports effects on job mobility and firm characteristics.

³²This effect is less directly tied to employer screening, since self-employed workers do not face an employer in the same way as wage workers. One interpretation is that certification may also operate as a broader market signal, helping self-employed workers communicate skills to clients, contractors, or other parties that hire their labor.

³³Appendix Table H.3 reports effects on job satisfaction, food insecurity, and anxiety.

9 Conclusions

This paper provides experimental evidence on whether a skill certificate can improve labor market outcomes for workers whose skills are difficult for employers to assess. We study a randomized offer of access to an occupation-specific certification program in Colombia, implemented among both locals and Venezuelan migrants. The program is designed to certify existing skills rather than provide training, making it a useful setting to study the value of labor market signals.

The effects differ across groups. For locals, the offer of access to the certification program increases employment and reduces inactivity. The earnings effect is driven by this extensive-margin response rather than by higher wages among those already employed. The dynamic estimates show that these effects emerge quickly and partly persist after one year. The evidence from the mechanism is consistent with certification reducing screening frictions and facilitating labor market entry.

For migrants, the main effect is not employment. The offer of access to the certification program reduces skill downgrading, with little evidence of an increase in overall employment or wages. This pattern suggests that the certificate helps some migrants move into occupations that better match their education rather than bringing more migrants into employment. The evidence on reservation wages is consistent with greater selectivity among unemployed migrants, but this interpretation is suggestive because it relies on post-treatment employment status.

These findings have two implications. First, credible certification of existing skills can affect labor market outcomes without changing skills. Second, its effects depend on the margins along which workers face frictions. For locals, it mainly increases employment. For migrants, it mainly improves match quality. This distinction matters for the design and evaluation of migrant integration policies. Policies that only measure employment may miss gains that occur through better occupational allocation.

Several limitations remain. The estimates come from one cohort in Colombia's Pacific region and cover outcomes up to twelve months after certification. The migrant sample is smaller than the local sample, which limits precision for some outcomes. The experiment also does not directly observe employer beliefs or hiring behavior, so the signaling interpretation remains indirect. Future work that measures employer responses to certified and non-certified applicants would help distinguish more directly between demand-side screening, worker search behavior, and other mechanisms.

References

- Abebe, G., Caria, A. S., Fafchamps, M., Falco, P., Franklin, S., and Quinn, S. (2021). Anonymity or Distance? Job Search and Labour Market Exclusion in a Growing African City. *The Review of Economic Studies*, 88(3):1279–1310.
- Anderson, M. L. (2008). Multiple inference and gender differences in the effects of early intervention: A reevaluation of the abecedarian, perry preschool, and early training projects. *Journal of the American Statistical Association*, 103(484):1481–1495.
- Bahar, D., Ibáñez, A. M., and Rozo, S. V. (2021). Give me your tired and your poor: Impact of a large-scale amnesty program for undocumented refugees. *Journal of Development Economics*, 151:102652.
- Bassi, V. and Nansamba, A. (2022). Screening and signaling non-cognitive skills: Experimental evidence from Uganda. *The Economic Journal*, 132:471–511.
- Battisti, M., Giesing, Y., and Laurentsyeve, N. (2019). Can job search assistance improve the labour market integration of refugees? evidence from a field experiment. *Labour Economics*, 61:101745.
- Busso, M., DiNardo, J., and McCrary, J. (2014). New evidence on the finite sample properties of propensity score reweighting and matching estimators. *The Review of Economics and Statistics*, 96(5):885–897.
- Busso, M., Montaña, S., and Muñoz-Morales, J. (2025). Signaling specific skills and the labor market of college graduates. *The Review of Economics and Statistics*, pages 1–46.
- Carranza, E., Garlick, R., Orlin, K., and Rankin, N. (2022). Job search and hiring with two-sided limited information about workseekers' skills. *American Economic Review*, forthcoming.
- Carranza, E. and McKenzie, D. (2024). Job training and job search assistance policies in developing countries. *Journal of Economic Perspectives*, 38(1):221–244.
- Corrales, J. (2015). The authoritarian resurgence: autocratic legalism in Venezuela. *Journal of Democracy*, 26(2):37–51.
- Dahlberg, M., Egebark, J., Vikman, U., and Özcan, G. (2024). Labor market integration of refugees: RCT evidence from an early intervention program in Sweden. *Journal of Economic Behavior & Organization*, 217:614–630.
- Delgado-Prieto, L. (2024). Immigration, wages, and employment under informal labor markets. *Journal of Population Economics*, 37(55):1432–1475.
- Delgado-Prieto, L. (2025). Firms, regularization, and the labor market integration of migrants. *Mimeo*.
- Dustmann, C., Frattini, T., and Preston, I. P. (2012). The Effect of Immigration along the Distribution of Wages. *The Review of Economic Studies*, 80(1):145–173.
- Eble, A. and Hu, F. (2022). Signals, Information, and the Value of College Names. *The Review of Economics and Statistics*, pages 1–45.

- Farber, H. S. and Gibbons, R. (1996). Learning and wage dynamics. *The Quarterly Journal of Economics*, 111(4):1007–1047.
- Faria, H. J. (2008). Hugo Chávez against the backdrop of Venezuelan economic and political history. *The Independent Review*, 12(4):519–535.
- Foged, M., Hasager, L., and Peri, G. (2024a). Comparing the effects of policies for the labor market integration of refugees. *Journal of Labor Economics*, 42(S1):S335–S377.
- Foged, M., Hasager, L., Peri, G., Arendt, J. N., and Bolvig, I. (2024b). Language training and refugees integration. *The Review of Economics and Statistics*, 106(4):1157–1166.
- Foged, M. and Van der Werf, C. (2023). Access to language training and the local integration of refugees. *Labour Economics*, 84:102366.
- Hainmueller, J., Lawrence, D., Gest, J., Hotard, M., Koslowski, R., and Laitin, D. (2018). A randomized controlled design reveals barriers to citizenship for low-income immigrants. *Proceedings of the National Academy of Sciences of the United States of America*, 115:939 – 944.
- Heller, S. B. and Kessler, J. B. (2021). Information frictions and skill signaling in the youth labor market. Working Paper 29579, National Bureau of Economic Research.
- Ibáñez, A. M., Moya, A., Ortega, M. A., Rozo, S. V., and Urbina, M. J. (2024). Life Out of the Shadows: The Impacts of Regularization Programs on the Lives of Forced Migrants*. *Journal of the European Economic Association*, page jvae044.
- ILO (2012). *International Standard Classification of Occupations: ISCO-08. Volume I: Structure, Group Definitions and Correspondence Tables*. International Labour Office, Geneva.
- Jaschke, P., Löbel, L.-M., Krieger, M., Legewie, N., Kroh, M., Jacobsen, J., and Schacht, D. (2022). Mentoring as a grassroots effort for integrating refugees evidence from a randomised field experiment. *Journal of Ethnic and Migration Studies*, 48(17):4085–4105.
- Joona, P. A. and Nekby, L. (2012). Intensive coaching of new immigrants: an evaluation based on random program assignment. *The Scandinavian Journal of Economics*, 114(2):575–600.
- Kaplan, D. S., Rivera, A. R., and van der Werf, C. (2023). A fair chance for migrants in the labor market.
- Kelley, E. M., Ksoll, C., and Magruder, J. (2024). How do digital platforms affect employment and job search? evidence from India. *Journal of Development Economics*, 166:103176.
- Landaud, F., Maurin, E., Willage, B., and Willen, A. (2024). The Value of a High School GPA. *The Review of Economics and Statistics*, pages 1–24.
- Lebow, J. (2023). Immigration and occupational downgrading in Colombia. *Journal of Development Economics*, pages 103–164.
- Lee, D. S. (2009). Training, Wages, and Sample Selection: Estimating Sharp Bounds on Treatment Effects. *The Review of Economic Studies*, 76(3):1071–1102.

- Leuven, E. and Oosterbeek, H. (2011). Overeducation and mismatch in the labor market. In *Handbook of the Economics of Education*, volume 4, pages 283–326. Elsevier.
- Liebig, T., Khoudour, D., and Harris, J. (2023). How do migrants fare in latin america and the caribbean?: Mapping socio-economic integration.
- Loiacono, F. and Silva-Vargas, M. (2023). Matching with the right attitude: The effect of matching firms with refugee workers. *London: PEDL*.
- MacLeod, B., Riehl, E., Saavedra, J., and Urquiola, M. (2017). The big sort: College reputation and labor market outcomes. *American Economic Journal: Applied Economics*, 9(3):223-261.
- McKenzie, D. (2017). How effective are active labor market policies in developing countries? a critical review of recent evidence. *The World Bank Research Observer*, 32(2):127–154.
- Pallais, A. (2014). Inefficient hiring in entry-level labor markets. *American Economic Review*, 104(11):3565–99.
- R4V (2024). América latina y el caribe: Refugiados y migrantes venezolanos en la región - noviembre 3. Accessed: 25 February 2025.
- Rodríguez, F. (2008). An empty revolution-the unfulfilled promises of Hugo Chavez. *Foreign Aff.*, 87:49.
- Spence, M. (1973). Job Market Signaling. *The Quarterly Journal of Economics*, 87(3):355–374.
- UNDESA (2025). International migrant stock 2024: Key facts and figures. Report UN DESA/POP/2024/DC/No. 13, United Nations Department of Economic and Social Affairs, Population Division, New York. Advance unedited version.

Appendices

Contents of the Appendix

A	Venezuelan Migration to Colombia	2
B	Skills Certification Program	4
B.1	Institutional Background on the Skill Certification Program	4
B.2	The “Skills Pay Off” Program	4
B.3	Program Roll-Out and the 2024 Cohort	7
B.4	Treatment Bundle of the 2024 Cohort	8
C	Outcomes	10
C.1	Pre-Specified Outcomes in the Pre-Analysis Plan and Definitions	10
C.2	Additional Details of the Skill-Downgrading Measure	12
C.3	Evaluation of Model Classification	13
D	Balance	17
E	Attrition Tests	18
F	Heterogeneity Tests	19
G	Robustness	20
G.1	Lee Bounds Estimations by Wave and Migrant Status	20
G.2	Skill Downgrade	23
G.3	Re-weighted Estimations	24
H	Secondary Outcomes	26

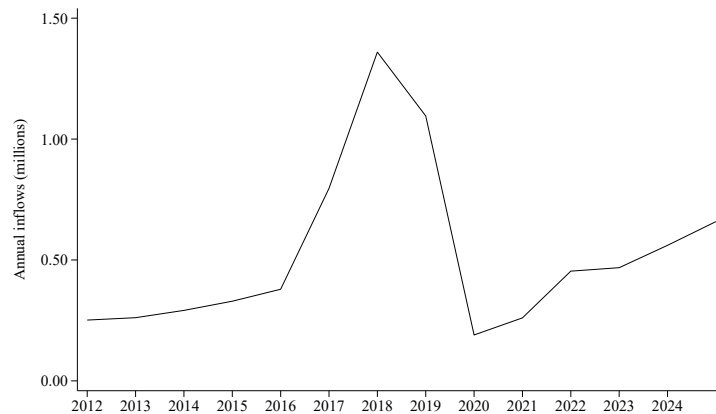
A Venezuelan Migration to Colombia

Forced displacement has increased in recent years, driven by armed conflict, political instability, economic crises, and natural disasters. Recent estimates indicate that more than 304 million people are currently displaced worldwide, including approximately 8 million asylum seekers. Latin America is an important part of these flows, with more than 48 million migrants recorded in 2024 (UNDESA, 2025).

A major driver of recent migration in Latin America is the economic and political crisis in Venezuela. Since the late 1990s, Venezuela has experienced increasing state intervention, fiscal expansion, institutional deterioration, and political repression (Rodríguez, 2008; Corrales, 2015; Faria, 2008). The economy remained highly dependent on oil revenues, leaving it exposed to the collapse in global oil prices in 2014. The subsequent deterioration contributed to hyperinflation, shortages of basic goods, and a severe humanitarian crisis.

Colombia became the main destination for Venezuelan migrants, partly because of geographic proximity and a shared language (Bahar et al., 2021). The two countries share a border of more than 2,000 kilometers. As of 2024, Colombia hosted approximately 2.86 million Venezuelan migrants, or roughly one-third of all Venezuelan migrants, with the bulk of the inflow concentrated between 2016 and 2020 (see Appendix Figure A.1; R4V, 2024). This inflow placed pressure on Colombia's labor market and institutional capacity (Delgado-Prieto, 2024).

Appendix Figure A.1: Annual Inflows of Venezuelans to Colombia



Notes: Annual entries of Venezuelans recorded at Colombian border crossings. Source: Colombian migration authority.

In response, Colombia implemented a large-scale regularization process that granted legal status to many Venezuelan migrants (Bahar et al., 2021; Ibáñez et al., 2024). As of August 2024, Colombia had issued over 2,092,619 temporary protection permits under the *Estatuto Temporal de Protección*, a program that allows migrants to work legally and access health and education services. As in the rest of the region, Venezuelan migrants in Colombia represent a demographic and economic asset: they are younger, participate in the labor force at higher rates, and are more likely to be employed than native workers (Kaplan et al., 2023). Yet this potential is offset by systematic disadvantages in job quality—migrants across LAC face higher rates of informality (47.8% versus 40.4% among natives) and overqualification (23.9% versus 17.0%), reflecting a broader pattern of human capital underutilization (Kaplan et al., 2023).

A key mechanism behind this gap is occupational mismatch. Venezuelan migrants continue to face high rates of informality ([Delgado-Prieto, 2025](#)), poverty, and precarious employment, particularly among those whose skills and credentials are difficult to verify in the Colombian labor market ([Bahar et al., 2021](#); [Lebow, 2023](#)). This reflects a broader regional pattern: across Latin America and the Caribbean, approximately 24% of highly educated foreign-born workers are employed in low- or medium-skilled jobs, compared to 17% of their native-born counterparts ([Kaplan et al., 2023](#)). The Colombian case has a distinct profile within this trend: Venezuelan migrants in Colombia tend to have lower levels of formal education than those who settled further afield—in Peru, Ecuador, or Chile—likely because migrants with higher educational attainment had the resources to travel longer distances ([Kaplan et al., 2023](#); [Liebig et al., 2023](#)). As a result, overqualification rates in Colombia, while present, are less pronounced than in those destinations. Yet even within this lower-education population, skill downgrading is prevalent, with many migrants working in occupations below their actual competency level ([Bahar et al., 2021](#); [Lebow, 2023](#)). These frictions—informality, credential barriers, and occupational mismatch—constitute the principal integration gaps that employment support policies have sought to address.

B Skills Certification Program

B.1 Institutional Background on the Skill Certification Program

The Colombian skill certification program, formally known in Spanish as “*Programa de Evaluación y Certificación de Competencias Laborales*”, was established under Decree 933 of 2003 and is administered by SENA (Servicio Nacional de Aprendizaje), the Colombian Vocational Training Agency. The program was created to formally recognize occupational skills acquired through work experience, on-the-job training, or non-formal learning, complementing SENA’s traditional role as the country’s main provider of technical and vocational training. Certification is occupation-specific and is based on national occupational standards (*Normas Sectoriales de Competencia Laboral*) developed in consultation with sectoral round-tables (*Mesas Sectoriales*) that bring together representatives from industry, workers’ organizations, government, and academia. This structure ensures that the competencies assessed reflect skills demanded in the Colombian labor market.

Since its inception, the program has expanded substantially in both scope and scale. It currently covers 2,405 competencies across manufacturing, agriculture, retail, and service occupations, and certifies more than 200,000 workers each year throughout the country. Certification is free of charge for applicants and is delivered by qualified assessors trained and accredited by SENA, who follow standardized evaluation protocols to ensure consistency across regions and competencies.

B.2 The “Skills Pay Off” Program

B.2.1 Origin and Eligibility

Saber Hacer Vale (“Skills Pay Off”) was introduced as part of Colombia’s National Development Plan 2018–2022 (*Plan Nacional de Desarrollo 2018–2022: Pacto Por la Equidad*), a government strategy aimed at promoting social inclusion and productivity. The program is implemented jointly by the Colombian Ministry of Labor and SENA. It preserves the core features of the underlying skill certification program described above, but complements it with additional light-touch resources and financial assistance (Appendix B.2.2 describes these components).

The program targets foreign migrants and Colombians classified as vulnerable. To be eligible for the additional subsidies, participants must meet at least one of the following criteria: being a household head or having dependents, being between 18 and 28 years old or over 60, belonging to an ethnic group or the LGBTIQ+ community, being a migrant or Colombian returnee,¹ or having low income.² At least 30 percent of the available slots are reserved for migrants with regular permit status.

Skills Pay Off is implemented in four stages. First, the Ministry of Labor identifies a geographic area for implementation and defines the set of competencies to be certified

¹Participants must be registered in the public list known as *Registro Único de Retornados*.

²Colombia classifies its population based on a score used to target social aid programs, known as SISBEN (*Sistema de Identificación de Potenciales Beneficiarios de Programas Sociales*). The first three categories in the SISBEN score are considered low income and eligible for social aid.

in consultation with productive-sector representatives.³ Second, the number of participants is determined according to the available budget. Third, the call for applications is launched, and candidates complete an enrollment form on the Ministry's website. Finally, applicants' documentation is reviewed and verified to determine eligibility for the program.

B.2.2 Treatment Bundle

Skills Pay Off complements the certification process with two additional components: (i) information and support resources, and (ii) financial assistance during participation.

Information and Support Resources. This component bundles three resources that distinguish *Skills Pay Off* from the underlying SENA certification:

1. Motivational support: The program provides participants with a tracking mechanism (using text messages, emails, phone calls, or in-person aid) intended to avoid dropout. This component is only offered to treated individuals who do not show up for the evaluation process. It applies only to participants who miss the evaluation appointment and therefore affects a small share of the treatment group.
2. Occupational support: The intervention includes a one-hour voluntary online training on using the digital platform *TuBio*, a digital tool that creates a curriculum template and provides recommendations on potential occupations. *TuBio* gathers information about the participant's prior education, experience, activities (such as hobbies, household chores, voluntary work, and others), languages, and competencies (see Figure B.1 for a list of the offered competencies). Based on this information, *TuBio* creates a curriculum template (an example is shown in Figure B.2) and provides feedback about potential occupations that might be of interest to the participant (Figure B.3 presents an example of recommended occupations for the profile shown in Figure B.2).

Appendix Figure B.1: Competencies Required by the Occupational Support

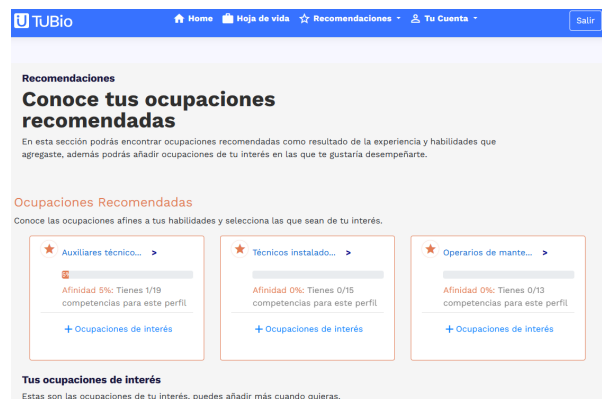
Mis competencias		
Has añadido 1 competencias		
Matemáticas	Servicio al cliente	Computadoras, electrónica y automatización
Telecomunicación	Manejo de las TIC	Mantenimiento
Evaluación y control de actividades	Trabajo en equipo	Pensamiento crítico
Comprensión de lectura	Escucha activa	Comunicación asertiva
Orientación al servicio	Análisis de control de calidad	Vigilancia de las operaciones
Mantenimiento de equipos	Manejo de imprevistos	Reparación

³The competencies evaluated are typically selected based on labor demand data and the historical success of technical certifications.

Appendix Figure B.2: Curriculum Created by the Occupational Support



Appendix Figure B.3: Occupational Recommendations Given by the Occupational Support



3. Labor Market Platforms Training: *Skills Pay Off* provides beneficiaries with information about the Public Employment Service. This mainly includes access to the public job-search platform. During the first session of the evaluation process, participants receive information about these platforms. If they request it, they can then receive support to register on these services.

Financial Incentives and Program Cost. All beneficiaries of *Skills Pay Off* receive food and connectivity subsidies totaling approximately \$82 USD. Some beneficiaries qualify for additional support, including \$75 USD for those with dependents, a \$30 USD transportation subsidy if the assessment is conducted in person, and a \$30 USD subsidy for those enrolled in the remedial session. Participants can receive up to \$217 USD in total support, equivalent to approximately 70 percent of the monthly minimum wage. The average cost per participant is approximately \$182 USD, as shown in Table B.1. This cost is lower than the cost of some other migrant labor market integration programs, such as the language training program for refugees studied by Foged et al. (2024b). The Colombian Ministry of Labor has faced budget constraints that may limit future implementation of the program. In a parallel project, we study a lower-cost certification intervention with fewer financial incentives, which costs approximately \$60 USD per participant and will allow us to assess scalability and cost effectiveness more directly.

Appendix Table B.1: Implementation Costs of *Skills Pay Off*

Year	Full Cost USD	Beneficiaries	Cost per-person USD
2021	373,253	3,050	122.4
2022	402,959	3,150	127.9
2023	357,129	2,500	142.9
2024	284,923	855	333.2
Avg. cost per person			181.6

Notes: Cost per-person USD is calculated as the ratio between total implementation costs and the number of beneficiaries in each year. The average cost per person is computed using the full set of yearly observations shown in the table.

B.3 Program Roll-Out and the 2024 Cohort

Before 2024, access to *Skills Pay Off* was allocated on a first-come, first-served basis. More than 12,000 individuals were certified through the program during its first four years, including approximately 4,000 migrants. These rounds were implemented nationwide and covered 38 competencies. Table B.2 reports participation by implementation wave since 2021.

Appendix Table B.2: Yearly Participation in *Skills Pay Off* Program

Call	Actual Beneficiaries	Colombian Migrants	% of Migrants	
2021	2,947	1,989	958	32.51
2022-1	1,636	1,114	522	31.91
2022-2	4,429	2,943	1,486	33.55
2023	2,623	1,581	1,042	39.73
2024	855	605	250	29.24
Total	12,490	8,232	4,258	

Notes: The table reports historical participation figures for *Skills Pay Off*.

The 2024 call was advertised through several channels, including social media, radio, and television. On-site agents were also hired to facilitate enrollment. Interested individuals registered through an online application form between June and August 2024. Submitted documents were reviewed between August 1st and September 3rd to determine whether applicants met the program's eligibility criteria.

The five competencies certified in the 2024 cohort were defined as follows. *Health services* referred to care activities performed according to protocols for basic daily activities and degree of autonomy. *Food handling* covered food handling according to technical procedures and regulations. *Food preparation* covered preparation based on production orders and standard recipes. *Design and tailoring* covered modifications according to tailoring and dressmaking techniques. *Recycling operations* covered conditioning recovered materials according to technical requirements. Table B.3 shows the number of participants by treatment status.

Appendix Table B.3: Sample sizes by Competencies and Treatment Status

Competency	Local Sample		Migrant Sample	
	Treatment	Control	Treatment	Control
Health Services	294	459	36	34
Food Preparation	153	300	132	122
Food handling	72	130	48	41
Design and tailoring	42	111	18	23
Recycling Operations	44	71	16	14
Total	605	1071	250	234

Notes: The table reports the distribution of participants across competencies, by treatment status and separately for locals and migrants.

B.4 Treatment Bundle of the 2024 Cohort

Although *Skills Pay Off* includes motivational support, occupational support, and platform training, it is unlikely that the main results are driven by these components. Three reasons support this interpretation. First, these resources are free and publicly available regardless of program status, and applicants in both the treatment and control

groups were informed of their availability at registration. Second, use of these resources is voluntary, including among individuals assigned to treatment, so any effects would depend on take-up by a self-selected subset of participants. Third, the resources are light-touch and unlikely, on their own, to generate large effects on the outcomes we study (Carranza and McKenzie, 2024; McKenzie, 2017; Kelley et al., 2024).

Appendix Table B.4 provides empirical support for this interpretation by reporting take-up of the *TuBio* online platform across treatment and control groups, separately for locals and migrants. Overall take-up is low: around 2 percent among locals and 5 percent among migrants. For locals, usage is statistically indistinguishable between the treatment and control groups (2.1 percent versus 1.9 percent; p -value = 0.709). For migrants, treatment-assigned participants are more likely to use *TuBio* than controls (8.4 percent versus 3.8 percent; p -value = 0.037), but take-up remains low in absolute terms in both groups, making it unlikely to drive the skill-downgrading effects documented among migrants.⁴

Appendix Table B.4: TuBio Usage by Treatment and Migration Status

	<i>Locals</i>			<i>Migrants</i>			
	Treatment	Control	p-value	Treatment	Control	p-value	Total
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Used TuBio	13	20	0.709	21	9	0.037	63
Percentage	2.1%	1.9%		8.4%	3.8%		2.9%
N	605	1,071		250	234		2,160

Notes: Each cell shows the number of participants who accessed the platform and the corresponding share of the group. Column (3) and (6) report the p-value from an OLS regression of platform access on treatment assignment, controlling for competence fixed effects, estimated separately for locals and migrants, and with robust standard errors.

Platform training consisted of a one-hour session on the Public Employment Service, focused on how to access the public job-search platform. This light-touch intervention is unlikely to drive the estimated effects.

Finally, only 4 percent of treatment-assigned participants withdrew before assessment, so motivational support was relevant for a small share of the treatment group (see Table 1). Taken together, these patterns reinforce the interpretation that the intervention primarily operated through access to certification rather than through additional support components.

⁴Data on use of the Public Employment Service are not available. However, the service is publicly accessible, and all participants were informed of its existence at registration.

C Outcomes

C.1 Pre-Specified Outcomes in the Pre-Analysis Plan and Definitions

Table C.1 reports the outcome measures used in the paper. For each outcome, the table gives the variable name, its definition, and whether it was included in the pre-analysis plan (PAP). Some reported outcomes are constructed from pre-specified variables to better align the analysis with the paper's research questions. Occupation and sector variables are used to construct measures of occupation change and sector change. Occupation is also used to construct the skill-downgrading measure. Inactivity is derived from employment status, and full-time work is derived from weekly hours worked.

There are three outcomes in the pre-analysis plan that we intended to collect and use but do not include in the analysis. First, we do not report maximum expected salary because baseline responses contained implausible values, raising concerns about measurement reliability. Second, we do not use perceived program impact in the main analysis because it was collected only for treated participants in the first follow-up. We report only the treated-group average for descriptive purposes. Third, we do not report time to first job offer because this item was not included in the follow-up questionnaires. The final survey instrument prioritized realized labor-market outcomes and contemporaneous job-search measures, given survey-length constraints and concerns about recall error in the timing of job offers.

Appendix Table C.1: Variable Definitions

Category	Variable	In PAP	Definition
Labor-Market	Employment status	Yes	Indicator equal to one if the respondent reports being employed in the reference week.
	Inactivity	No	Indicator equal to one if the respondent is neither employed nor actively searching for a job.
	Weekly hours worked	Yes	Self-reported weekly hours worked in the main job; set to zero for non-employed respondents.
	Full-time work	No	Indicator equal to one if the respondent reports working seven or more hours per day.
	Log monthly wage	Yes	Log of monthly labor earnings from the main job in USD, computed as $\log(\text{wage} + 1)$ with zero earnings imputed for non-employed respondents.
	Formal employment	Yes	Indicator equal to one if the respondent is employed and contributes to the pension system.
	Skill downgrade	Yes	Indicator equal to one if a respondent with post-secondary education is employed in an occupation at skill level 1 or 2 of the Colombian Classification of Occupations (CUOC).
Job Search	Search intensity	No	Self-reported weekly hours spent on job-search activities by the unemployed.
	Unemployment duration	Yes	Self-reported duration of the current job-search spell in months, top-coded at five or more.

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Table C.1 continued from previous page

Category	Variable	In PAP	Definition
	Log reservation wage	Yes	Natural logarithm of the monthly reservation wage in USD. The minimum monthly salary at which the respondent would accept a job (unemployed) or switch jobs (employed).
	Maximum expected salary	Yes	Self-reported maximum monthly salary the respondent expects to earn given their qualifications, in USD. Collected only at baseline.
Education and Training	Enrollment in additional education (any)	Yes	Indicator equal to one if the respondent reports enrollment in or completion of any additional educational activity in the prior three months.
	Enrollment in certification program	Yes	Indicator equal to one if the respondent reports being enrolled in a different skill-certification program.
	Enrollment in training course	Yes	Indicator equal to one if the respondent reports being enrolled in a short training course or diploma program.
	Enrollment in degree program	Yes	Indicator equal to one if the respondent reports enrollment in a professional, technological, or technical career program.
Firm Characteristics and Job Mobility	Solo / self-employed	No	Indicator equal to one if the respondent works alone in a single-person firm.
	Small firm	Yes	Indicator equal to one if the respondent is employed at a firm with 2–10 workers.
	Medium/large firm	Yes	Indicator equal to one if the respondent is employed at a firm with 11 or more workers.
	Economic sector of employer	Yes	Indicators for the broad economic sector of the respondent's main employer, approximating the 1-digit level of the industry classification (CIIU).
	Changed sector	No	Indicator equal to one if the sector of the respondent's employer at the follow-up differs from the sector reported at baseline (conditional on being employed at both points).
	Occupation type	Yes	Indicators for the broad occupation groups (1-digit CUOC) of the respondent's main job.
	Changed job	No	Indicator equal to one if a respondent employed at both baseline and the follow-up holds a job that began after baseline.
	Changed Occupation	No	Indicator equal to one if the broad occupation group (1-digit CUOC) of the main job differs between baseline and follow-up, conditional on being employed at both.
	Got a job offer	Yes	Indicator equal to one if the respondent reports having received at least one job offer in the reference period.
	Number of job offers	Yes	Self-reported count of job offers received in the reference period; set to zero if the respondent reports no offers.
	Time to first job offer	Yes	Time elapsed between starting the job search and the first job offer.

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Table C.1 continued from previous page

Category	Variable	In PAP	Definition
Job Satisfaction and Wellbeing	Wants a different job	Yes	Indicator equal to one if an employed respondent reports wanting to change jobs.
	Job satisfaction	Yes	Indicator equal to one if the respondent reports being “satisfied” or “very satisfied” with their current job, on a five-point satisfaction scale. Collected at follow-ups 2 and 3.
	Anxiety / nervousness	Yes	Indicator equal to one if the respondent’s GAD-7 total score indicates moderate or severe anxiety (score ≥ 10 out of 21).
	GAD-7 anxiety score	Yes	Total score on the Generalized Anxiety Disorder 7-item scale, ranging from 0 to 21, with higher values indicating more severe symptoms.
	Skipping a meal	Yes	Indicator equal to one if the respondent reports having skipped a meal due to lack of economic resources “sometimes” or “often” in the reference period.
Stated Preferences about Program	WTP for certification (USD midpoint)	Yes	Willingness-to-pay measure in USD, constructed as the midpoint of the reported willingness-to-pay interval. Collected only at follow-up 1 and 3.
	Willingness to pay for certification	Yes	Indicator equal to one if the respondent reports being willing to pay at least 50,000 COP (approximately 12 USD) for the skill certification. Collected only in follow-up 1 and 3.
	Willingness to participate without subsidies	Yes	Indicator equal to one if the respondent reports being “willing” or “very willing” to have participated in the program without financial subsidies. Collected only at follow-up 1.
	Perceived program impact	Yes	Indicator equal to one if the respondent perceives the program as having helped them improve their employment situation “moderately” or “a lot.” Collected only at follow-up 1 only for treated participants.

Notes: This table lists the outcomes and their definitions. Monetary outcomes are reported in USD using the average official Colombian peso exchange rate in 2024 of 4052.27 COP/USD.

C.2 Additional Details of the Skill-Downgrading Measure

We asked each participant to report their occupation title and describe the tasks performed in that job. Surveyors were instructed to assign an occupation code based on this information.⁵ Codes were assigned using the categories in Table C.2. Occupation codes 5 (Service and Sales) and 9 (Elementary Occupations) were further disaggregated into 3-digit subgroups because they contained the largest number of respondents at baseline, allowing for a more precise manual classification.

Figure C.1 illustrates the procedure used to match each job description to a CUOC occupation code. First, we provide the job title and task description to the OpenAI o4-mini model and ask the model to assign a 5-digit CUOC code. To improve precision,

⁵CUOC codes follow a hierarchical structure: 1-digit macro-occupations, 2-digit main subgroups, 3-digit subgroups, 4-digit primary occupations, and 5-digit occupation categories. Surveyors received a simplified list covering all ten macro-occupations.

we restrict the set of possible 5-digit codes using the occupation category manually assigned during the survey. We also provide the full set of 3-digit codes to allow for potential surveyor misclassification. When the procedure identifies a mismatch between the manually assigned category and the model's classification, we repeat the assignment using the corrected occupation category to obtain the final 5-digit code.

C.3 Evaluation of Model Classification

To assess the quality of the matching procedure, two research assistants independently evaluated a random sample of 200 model-generated matches. For each observation, they assigned one of three labels: (1) Good, (2) Partial, or (3) No match. A match was labeled "Good" when the assigned occupation was the best available classification. It was labeled "Partial" when the assigned occupation was related to the job description but a more precise alternative existed, such as a heavy-truck driver classified as a bus driver. It was labeled "No match" when the assigned occupation did not reflect the job description. For observations labeled "Partial" or "No match", the research assistants also assigned a manual 5-digit CUOC code.

Table C.3 reports the results of the validation exercise. The two human annotators agreed on 87% of classifications. We then consolidated their annotations using a conservative adjudication rule: when labels differed, the stricter classification prevailed, with "No match" ranked below "Partial" match and "Partial" match ranked below "Good" match. Under this rule, 76% of model-generated matches were classified as "Good" (Wilson CI: [69.6%, 81.4%]), and 95% were classified as either "Good" or "Partial" (Wilson CI: [91.0%, 97.3%]).

To assess the sensitivity of the *skill-downgrading* measure, we use the original model-generated codes for observations labeled "Good" and the research assistants' manual annotations for observations labeled "Partial" or "No match". When the research assistants assigned different 5-digit occupation codes, they deliberated and agreed on a final code. We then map these human-validated occupation codes to skill levels using the classification provided by the Colombian statistical office (DANE).⁶

Our *skill-downgrading* estimates would be affected by occupation-matching errors only if the model-generated and human-assigned occupations imply different skill levels. Table C.5 reports this comparison. We group occupations into two skill categories: those requiring secondary education or less (levels 1 and 2) and those requiring post-secondary education (levels 3 and 4) (ILO, 2012). Agreement between the human and model classifications across these two groups is 97.5%. A McNemar test yields a *p-value* of 0.375, indicating no statistically significant difference between the two classifications. This suggests that even when the model does not identify the exact occupation, it usually assigns an occupation with the same broad skill level. This may partly reflect the algorithm's use of the manually assigned occupation category

⁶Skill levels are defined by the International Labor Organization (ILO) according to the complexity of tasks performed in each occupation. Skill Level 1 is associated with simple, routine physical or manual tasks and typically requires primary education. Skill Level 2 covers tasks such as operating machinery, driving vehicles, and performing maintenance and repair, which typically require secondary education. Skill Level 3 encompasses complex technical and practical tasks that require specialized knowledge in a particular field. Skill Level 4 involves tasks requiring advanced problem-solving, decision-making, and creativity, usually associated with a degree that takes three to six years to complete.

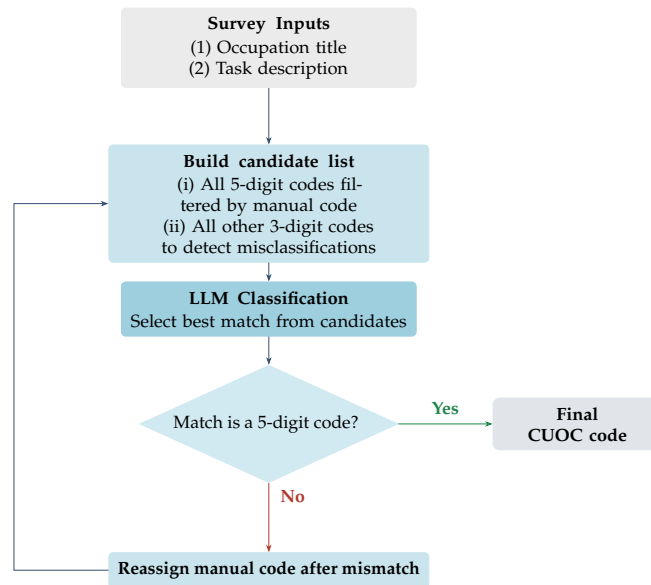
to restrict the candidate set, making the *skill-downgrading* measure less sensitive to fine-grained occupation-coding errors.

Appendix Table C.2: Occupation Categories Used in the Surveys

CUOC Code	Occupation Name
Panel A: Major Groups (1-digit)	
0	Armed Forces Occupations
1	Managers
2	Professionals
3	Technicians and Associate Professionals
4	Clerical Support Workers
6	Skilled Agricultural, Forestry and Fishery Workers
7	Craft and Related Trades Workers
8	Plant and Machine Operators and Assemblers
Panel B: Occupation Sub-Groups (3-digit)	
511	Travel Attendants and Travel Stewards
512	Cooks
513	Waiters and Bartenders
514	Hairdressers, Beauticians and Related Workers
515	Building and Housekeeping Supervisors
516	Other Personal Services Workers
521	Street and Market Salespersons
522	Shop Salespersons
523	Cashiers and Ticket Clerks
524	Other Sales Workers
531	Child Care Workers and Teachers' Aides
532	Personal Care Workers in Health Services
541	Protective Services Workers
911	Domestic, Hotel and Office Cleaners and Helpers
912	Vehicle, Window, Laundry and Other Hand Cleaning Workers
921	Agricultural, Forestry and Fishery Labourers
931	Mining and Construction Labourers
932	Manufacturing Labourers
933	Transport and Storage Labourers
941	Food Preparation Assistants
951	Street and Related Service Workers
952	Street Vendors (Excluding Food)
961	Refuse Workers
962	Other Elementary Workers

Notes: CUOC codes correspond to the Colombian Unified Classification of Occupations (ISCO-08 adaptation). Panel A reports 1-digit major groups used. Panel B reports 3-digit sub-major groups separation for the most frequently reported occupations.

Appendix Figure C.1: Two-Step LLM Classification Procedure



Notes: This figure describes the two-step procedure used to classify each participant's occupation into a 5-digit CUOC code.

Appendix Figure C.2: LLM Prompt for Occupation Classification

System message

You are a helpful assistant with deep knowledge of Latin-American labor markets. Avoid hallucinations. Also, you will have a list of occupation tuples, take that as ground truth for the query. Reply ONLY with the chosen tuple.

List of occupation tuples: «[Input list of occupation tuples](#)».

User prompt

You have three inputs in Spanish:

1. An occupation title: «[Input occupation title](#)».
2. An occupation description: «[Input occupation description](#)».
3. A list of occupation tuples: [Provided in the system message for this request]. Each tuple has the form ("code", "occupation name").

Task:

From the provided list of standardized occupation tuples, choose the one that best matches the given occupation title and description. Do not include any additional text, context, or chat.

Notes: Prompt used within the OpenAI API (o4-mini model) to classify each participant's occupation into a 5-digit CUOC code. The system message is shared across all API calls within the same occupation family. Inputs replaced before each call include: «[Input list of occupation tuples](#)», which contains the set of candidate CUOC codes; «[Input occupation title](#)», the job title reported by the participant; and «[Input occupation description](#)», the description of tasks reported by the participant.

Appendix Table C.3: Annotator Agreement on CUOC Codes

Annotator 1	Annotator 2		
	Good Match	Partial Match	No Match
Good Match	152	19	0
Partial Match	6	13	1
No Match	0	0	9

Notes: Annotator 1 versus Annotator 2 response distribution. Percentage of agreement of 87%.

Appendix Table C.4: Accuracy of LLM-Assigned CUOC Codes

Evaluation	N	%	95% CI
Good Match	152	76.0	[69.6, 81.4]
Partial Match	38	19.0	[14.2, 25.0]
Good + Partial Match	190	95.0	[91.0, 97.3]
No Match	10	5.0	[2.7, 9.0]
Total	200	100.0	

Notes: Accuracy using strictest label between two independent annotators. 95% Wilson confidence intervals in brackets.

Appendix Table C.5: Skill-Level Agreement Between Model and Human classification

LLM response	Human Label		Total
	Secondary or less	Post-Secondary	
Secondary or less (Levels 1-2)	168	4	172
Post-Secondary (Levels 3-4)	1	27	28
Total	169	31	200

Notes: Skill levels from CUOC classification grouped as Secondary Education (Skill Levels 1–2) and Post-Secondary (Skill Levels 3–4). Percentage of Agreement = 97.5%. McNemar’s test p -value = 0.375.

D Balance

Appendix Table D.1: Balance within Locals and Migrants

Variable	Locals			Migrants			Total		
	Treatment (1)	Control (2)	<i>p-value</i> (3)	Treatment (4)	Control (5)	<i>p-value</i> (6)	Locals (7)	Migrants (8)	<i>p-value</i> (9)
Panel A: General characteristics									
Demographics									
Age	42.671	42.350	0.819	35.964	36.073	0.947	42.466	36.017	0.000
Male	0.119	0.118	0.908	0.176	0.137	0.243	0.118	0.157	0.361
Caregiver (young or elderly)	0.526	0.551	0.111	0.464	0.449	0.811	0.542	0.457	0.441
No children under 5 years *	0.615	0.612	0.967	0.504	0.530	0.554	0.613	0.517	0.001
One child under 5 years *	0.233	0.223	0.592	0.348	0.338	0.804	0.227	0.343	0.000
More than one child under 5 years *	0.152	0.165	0.500	0.148	0.132	0.606	0.161	0.140	0.145
Head of household	0.539	0.533	0.925	0.628	0.679	0.207	0.535	0.653	0.000
Identifies as LGBTIQ	0.041	0.039	0.829	0.064	0.081	0.500	0.040	0.073	0.006
Disability (mental or physical)	0.058	0.057	0.997	0.028	0.060	0.104	0.057	0.043	0.233
Victim of conflict	0.448	0.479	0.283	0.021	0.031	0.524	0.468	0.026	0.000
Ethnic minority	0.326	0.369	0.100	0.032	0.030	0.890	0.353	0.031	0.000
Eligible for government assistance	0.618	0.614	0.705	0.548	0.568	0.641	0.616	0.558	0.015
Receives government or NGO assistance	0.140	0.140	0.781	0.012	0.013	0.913	0.140	0.012	0.000
Education									
Primary or less	0.198	0.187	0.551	0.080	0.051	0.209	0.191	0.066	0.000
Secondary	0.503	0.482	0.356	0.608	0.611	0.979	0.490	0.610	0.000
Technical education	0.217	0.246	0.139	0.136	0.150	0.643	0.236	0.143	0.000
University or postgraduate	0.082	0.084	0.783	0.176	0.188	0.721	0.083	0.182	0.000
Labor									
Employed	0.324	0.332	0.929	0.540	0.504	0.402	0.329	0.523	0.000
Pooled monthly salary (USD)	105.848	97.962	0.380	109.555	120.080	0.514	100.804	114.713	0.695
Less than 1 year of experience	0.145	0.127	0.347	0.100	0.094	0.850	0.134	0.097	0.097
1-2 years of experience	0.185	0.183	0.820	0.260	0.282	0.619	0.184	0.271	0.008
3-5 years of experience	0.202	0.193	0.506	0.268	0.295	0.489	0.196	0.281	0.006
6-10 years of experience	0.155	0.169	0.465	0.176	0.141	0.305	0.164	0.159	0.592
More than 10 years of experience	0.312	0.328	0.380	0.196	0.188	0.788	0.322	0.192	0.000
N (Sample Size)	605	1,071		250	234		1,676	484	
Panel B: Employed									
Monthly salary (USD)	140.053	142.248	0.828	138.463	159.594	0.470	141.467	148.319	0.781
Overall skill downgrading *	0.218	0.234	0.658	0.247	0.246	0.997	0.228	0.247	0.641
Skill downgrading among professionals *	0.053	0.024	0.230	0.111	0.130	0.662	0.035	0.120	0.003
Have written employment contract	0.138	0.134	0.998	0.086	0.085	0.890	0.136	0.086	0.055
Contributes to pension fund	0.179	0.190	0.686	0.078	0.094	0.660	0.186	0.085	0.000
Full-time job	0.510	0.451	0.186	0.437	0.500	0.391	0.472	0.466	0.614
Half-time job	0.301	0.321	0.645	0.296	0.288	0.614	0.314	0.292	0.440
Quarter-time job	0.189	0.228	0.265	0.267	0.212	0.621	0.214	0.241	0.157
N (Sample Size)	196	356		135	118		552	253	
Panel C: Unemployed									
Last job monthly salary (USD)	84.292	69.115	0.248	68.476	74.812	0.794	74.629	71.772	0.337
Reservation Wage (USD) *	347.416	355.453	0.533	324.837	361.976	0.137	352.661	342.041	0.430
Never been employed	0.032	0.011	0.040	0.026	0.017	0.677	0.019	0.022	0.478
Actively seeking job (last 4 weeks)	0.343	0.302	0.121	0.365	0.422	0.400	0.317	0.394	0.186
Job search intensity: less than 5 hrs/day	0.600	0.581	0.747	0.667	0.490	0.204	0.589	0.571	0.736
Job search intensity: 5-10 hrs/day	0.314	0.330	0.765	0.262	0.367	0.745	0.324	0.319	0.966
Job search intensity: Over 10 hrs/day	0.086	0.088	0.946	0.071	0.143	0.130	0.087	0.110	0.618
N (Sample Size)	409	715		115	116		1,124	231	

Note: Panel A covers all registered participants; Panels B and C restrict the sample to those employed and unemployed at baseline, respectively. Columns (1)–(2) and (4)–(5) report mean values by treatment assignment within locals and migrants. Columns (7)–(8) report mean values pooling treatment and control within each group. *P-values* in columns (3) and (6) test the null of equal means between treatment and control, estimated via OLS of the variable on a treatment indicator with competency-level fixed effects and robust standard errors, separately for locals and migrants. *P-values* in column (9) test the null of equal means between locals and migrants, estimated analogously with the migrant indicator as the regressor. Pooled salary combines current monthly salary for employed individuals with last monthly salary for unemployed individuals. Variables marked with * were collected in a complementary baseline survey administered three months after initial enrollment but prior to certification.

E Attrition Tests

Appendix Table E.1: Attrition and Survey Completion Rates across Follow-up Rounds

	Total (n)	Comp. Baseline (n) (%)	3 Months (n) (%)	6 Months (n) (%)	12 Months (n) (%)	Answered all rounds (n) (%)	Registration only (n) (%)
<i>Panel A: Locals</i>							
Treatment	605	540 89.26	564 93.22	537 88.76	515 85.12	492 81.32	28 4.63
Control	1071	948 88.52	974 90.94	933 87.11	881 82.26	835 77.96	52 4.86
<i>Panel B: Migrants</i>							
Treatment	250	236 94.40	238 95.20	222 88.80	207 82.80	201 80.40	4 1.60
Control	234	216 92.31	204 87.18	193 82.48	179 76.50	174 74.36	10 4.27
N	2160	1940 89.81	1980 91.67	1885 87.27	1782 82.50	1702 78.80	94 4.35

Notes: This table reports response rates at baseline and all follow-up rounds, disaggregated by nationality (Colombians vs. migrants) and treatment assignment (treatment vs. control). Columns show the number and percentage of participants who completed each survey round, answered all surveys, or responded only to the registration form without completing subsequent surveys (Reg. Only). Percentages are calculated relative to the original registration sample. These figures provide an overview of attrition patterns and survey completeness across experimental groups.

Appendix Table E.2: Differential Attrition by Follow-up Wave

	Comp. Baseline		3 months		6 months		12 months	
	Local	Migrant	Local	Migrant	Local	Migrant	Local	Migrant
Treatment	-0.007 (0.016)	-0.020 (0.023)	-0.025 (0.014)	-0.079*** (0.026)	-0.018 (0.017)	-0.064** (0.032)	-0.030 (0.019)	-0.065* (0.036)
Constant	0.115*** (0.010)	0.077*** (0.017)	0.091*** (0.009)	0.128*** (0.022)	0.129*** (0.010)	0.175*** (0.025)	0.178*** (0.012)	0.236*** (0.027)
Observations	1676	484	1676	484	1676	484	1676	484

Notes: Each column reports estimates from a linear probability model where the dependent variable equals one if the participant did not complete the survey in the corresponding wave and zero otherwise. *Treated* is an indicator for assignment to the certification program. All specifications include technical competence fixed effects. Regressions are estimated separately by migration status. Robust standard errors in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

F Heterogeneity Tests

Appendix Table F.1: Effects of Access to Skills Certification on Labor Market Outcomes

Intention To Treat Estimated using Equation (1)–Heterogeneity by Migrant Status

	Employed (1)	Inactive (2)	Outcomes Replaced by Zero if Non-Employed				
			Log Wage (3)	Formal emp. (4)	Weekly working hours (5)	Full time (7+ hrs) (6)	Skill Downgrade (7)
Treatment	0.057*** (0.021)	-0.035* (0.018)	0.740*** (0.283)	-0.008 (0.015)	2.033** (0.858)	0.037** (0.017)	0.003 (0.016)
Migrant	0.190*** (0.029)	-0.118*** (0.021)	2.789*** (0.401)	-0.046** (0.021)	10.126*** (1.502)	0.187*** (0.030)	0.114*** (0.030)
Treatment × Migrant	-0.089** (0.041)	0.047 (0.030)	-1.070* (0.562)	0.031 (0.029)	-3.476* (2.023)	-0.057 (0.041)	-0.083** (0.039)
P-value (T × Migrant)	0.028	0.120	0.057	0.297	0.086	0.165	0.034
Observations	5,647	5,647	5,634	5,647	5,647	5,647	5,647
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneity in pooled ITT effects by migrant status, estimated from a specification interacting treatment with migrant status. Assigned to treatment reports the ITT effect for locals ($M=0$). Migrant reports the baseline difference between migrants and locals. Treatment × Migrant reports the differential effect for migrants relative to locals. P-value (T × Migrant) reports the p-value for the test that the interaction term equals zero. Outcomes in columns (3)–(7) are replaced by zero for non-employed individuals. For non-employed individuals, monthly earnings are set to zero prior to the transformation $\log(\text{wage} + 1)$. All specifications include competence and wave fixed effects. Standard errors (in parentheses) are clustered at the individual level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

G Robustness

G.1 Lee Bounds Estimations by Wave and Migrant Status

Appendix Table G.1.1: Effects of Access to Skills Certification (ITT)

Lee Bounds – by follow-up Round using Full Sample

		Outcomes Replaced by Zero if Non-Employed					
	Employed	Inactive	Log Wage	Weekly working hours	Formal emp.	Full time (7+ hrs)	Skill Downgrade
<i>Panel A: 3 months</i>							
Treatment	0.046** (0.023)	-0.041** (0.019)	0.667** (0.304)	2.399** (0.998)	0.002 (0.014)	0.046** (0.020)	-0.018 (0.016)
Observations	1,980	1,980	1,980	1,980	1,980	1,980	1,980
Lower Bound	0.029	-0.072***	0.409	0.073	-0.034**	0.020	-0.053***
Upper Bound	0.067***	-0.034**	0.667**	2.399**	0.007	0.058***	-0.012
Mean Control	0.497	0.244	6.187	13.53	0.113	0.227	0.148
<i>Panel B: 6 months</i>							
Treatment	0.027 (0.023)	-0.012 (0.020)	0.329 (0.316)	0.947 (1.003)	-0.004 (0.015)	-0.000 (0.021)	-0.007 (0.017)
Observations	1,885	1,885	1,872	1,885	1,885	1,885	1,885
Lower Bound	0.016	-0.034*	0.124	-0.524	-0.028*	-0.020	-0.030*
Upper Bound	0.043*	-0.005	0.329	0.947	-0.000	0.008	-0.002
Mean Control	0.539	0.245	6.945	14.65	0.125	0.264	0.149
<i>Panel C: 12 months</i>							
Treatment	0.035 (0.024)	-0.017 (0.020)	0.464 (0.323)	0.213 (1.052)	-0.000 (0.017)	0.025 (0.022)	-0.024 (0.018)
Observations	1,782	1,782	1,782	1,782	1,782	1,782	1,782
Lower Bound	0.019	-0.049**	0.129	-1.823**	-0.034**	-0.000	-0.056***
Upper Bound	0.059**	-0.010	0.464*	0.213	0.006	0.037*	-0.018
Mean Control	0.557	0.232	7.120	16.21	0.147	0.280	0.172
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. Each panel presents results for a separate follow-up round (Panel A: 3 months, Panel B: 6 months, Panel C: 12 months). Outcomes in columns (3)–(7) are replaced by zero if the outcome is missing for non-employed individuals; a dummy for missing values is included as an additional control. Lower and upper bounds follow [Lee \(2009\)](#) and are estimated by trimming the group with excess respondents. We report Lee bounds for primary labor market outcomes only; mechanism outcomes in Table 2 are secondary and subject to the same attrition patterns documented here. Robust standard errors for the ITT coefficient are reported in parentheses. Standard errors for the bounds are obtained via bootstrap with 100 replications. All specifications include competence fixed effects and control for migrant status. Mean Control reports the control-group mean among observed respondents in the corresponding wave. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table G.1.2: Effects of Access to Skills Certification (ITT)

Lee Bounds – by follow-up Round using Sample of Locals

Outcomes Replaced by Zero if Non-Employed							
	Employed	Inactive	Log Wage	Weekly working hours	Formal emp.	Full time (7+ hrs)	Skill Downgrade
<i>Panel A: 3 months</i>							
Treatment	0.070*** (0.026)	-0.060*** (0.022)	0.938*** (0.348)	3.528*** (1.113)	-0.014 (0.017)	0.071*** (0.022)	0.002 (0.018)
Observations	1,538	1,538	1,538	1,538	1,538	1,538	1,538
Lower Bound	0.060**	-0.078***	0.818**	1.659*	-0.035**	0.054***	-0.018
Upper Bound	0.082***	-0.055**	0.938***	3.528***	-0.011	0.078***	0.006
Mean Control	0.461	0.270	5.683	11.78	0.123	0.190	0.125
<i>Panel B: 6 months</i>							
Treatment	0.046* (0.027)	-0.019 (0.024)	0.582 (0.363)	1.617 (1.090)	-0.003 (0.018)	0.009 (0.023)	0.021 (0.018)
Observations	1,470	1,470	1,470	1,470	1,470	1,470	1,470
Lower Bound	0.038	-0.032	0.439	0.280	-0.017	-0.004	0.006
Upper Bound	0.054*	-0.015	0.582	1.617*	-0.000	0.013	0.024
Mean Control	0.501	0.270	6.313	12.75	0.129	0.229	0.114
<i>Panel C: 12 months</i>							
Treatment	0.054** (0.027)	-0.024 (0.024)	0.694* (0.371)	0.777 (1.139)	-0.006 (0.020)	0.030 (0.024)	-0.008 (0.020)
Observations	1,396	1,396	1,396	1,396	1,396	1,396	1,396
Lower Bound	0.040	-0.050**	0.503	-1.516	-0.034**	0.006	-0.036**
Upper Bound	0.074**	-0.017	0.694*	0.777	-0.001	0.040	-0.003
Mean Control	0.522	0.251	6.592	14.23	0.157	0.247	0.152
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Assigned to certification (ITT) is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. Each panel presents results for a separate follow-up round (Panel A: 3 months, Panel B: 6 months, Panel C: 12 months). Outcomes in columns (3)–(7) are replaced by zero if the outcome is missing for non-employed individuals; a dummy for missing values is included as an additional control. Lower and upper bounds follow Lee (2009) and are estimated by trimming the group with excess respondents. We report Lee bounds for primary labor market outcomes only; mechanism outcomes in Table 2 are secondary and subject to the same attrition patterns documented here. Robust standard errors for the ITT coefficient are reported in parentheses. Standard errors for the bounds are obtained via bootstrap with 100 replications. All specifications include competence fixed effects. Mean Control reports the control-group mean among observed respondents in the corresponding wave. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table G.1.3: Effects of Access to Skills Certification (ITT)

Lee Bounds – by follow-up Round using Migrants Sample

Outcomes Replaced by Zero if Non-Employed							
	Employed	Inactive	Log Wage	Weekly working hours	Formal emp.	Full time (7+ hrs)	Skill Downgrade
<i>Panel A: 3 months</i>							
Treatment	-0.036 (0.045)	0.022 (0.032)	-0.228 (0.627)	-1.278 (2.226)	0.054** (0.027)	-0.036 (0.046)	-0.101*** (0.036)
Observations	442	442	442	442	442	442	442
Lower Bound	-0.071	-0.058**	-0.927	-4.843**	-0.027	-0.094**	-0.180***
Upper Bound	0.022	0.036	-0.228	-1.278	0.063***	-0.001	-0.090**
Mean Control	0.667	0.118	8.593	21.88	0.0637	0.402	0.221
<i>Panel B: 6 months</i>							
Treatment	-0.032 (0.045)	0.008 (0.033)	-0.432 (0.630)	-1.183 (2.378)	-0.005 (0.030)	-0.026 (0.049)	-0.048 (0.039)
Observations	415	415	415	415	415	415	415
Lower Bound	-0.053	-0.057	-0.833	-4.740*	-0.068***	-0.069	-0.108***
Upper Bound	0.018	0.016	-0.432	-1.183	0.003	0.003	-0.035
Mean Control	0.725	0.124	9.678	23.80	0.109	0.430	0.218
<i>Panel C: 12 months</i>							
Treatment	-0.030 (0.046)	0.001 (0.036)	-0.311 (0.649)	-2.015 (2.504)	0.019 (0.032)	0.005 (0.051)	-0.076* (0.043)
Observations	386	386	386	386	386	386	386
Lower Bound	-0.052	-0.067**	-0.700	-4.875**	-0.049*	-0.038	-0.139***
Upper Bound	0.025	0.012	-0.311	-2.015	0.028	0.040	-0.063
Mean Control	0.726	0.140	9.719	25.96	0.101	0.441	0.268
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. Each panel presents results for a separate follow-up round (Panel A: 3 months, Panel B: 6 months, Panel C: 12 months). Outcomes in columns (3)–(7) are replaced by zero if the outcome is missing for non-employed individuals; a dummy for missing values is included as an additional control. Lower and upper bounds follow Lee (2009) and are estimated by trimming the group with excess respondents. We report Lee bounds for primary labor market outcomes only; mechanism outcomes in Table 2 are secondary and subject to the same attrition patterns documented here. Robust standard errors for the ITT coefficient are reported in parentheses. Standard errors for the bounds are obtained via bootstrap with 100 replications. All specifications include competence fixed effects. Mean Control reports the control-group mean among observed respondents in the corresponding wave. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

G.2 Skill Downgrade

Appendix Table G.2.1: Effects of Access to Skills Certification on Alternative Measure of Skill Downgrading

Intention To Treat Effects estimated using Equation (1)

	Skill Downgrade
<i>Panel A: Full Sample</i>	
Treatment	-0.012 (0.013)
Mean Control	0.127
Observations	5647
<i>Panel B: Locals</i>	
Treatment	0.003 (0.014)
Mean Control	0.109
Observations	4404
<i>Panel C: Migrants</i>	
Treatment	-0.061* (0.033)
Mean Control	0.212
Observations	1243
Competence FE	Yes
Wave FE	Yes

Notes: The table reports pooled ITT estimates from Equation 1 for an alternative skill-downgrade outcome based on the official Colombian occupation codes (GEIH/CUOC) rather than the classification used in the main-text measure. Panels A, B, and C report estimates for the full sample, locals, and migrants, respectively. The outcome is set to zero for individuals not employed at follow-up. All specifications include competence and wave fixed effects. Full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Mean Control reports the control-group mean in the corresponding estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

G.3 Re-weighted Estimations

Appendix Table G.3.1: Covariate balance before and after propensity score reweighting

	Locals	Migrants	Migrants (normalized weights)	p-value	p-value
	(1)	(2)	(3)	(2) – (1)	(3) – (1)
<i>Panel A: Demographics</i>					
Age	42.47	36.02	41.09	0.00	0.25
Male	0.12	0.16	0.13	0.36	0.69
Household head	0.54	0.65	0.60	0.00	0.09
Has children	0.28	0.42	0.29	0.00	0.66
Caregiver	0.54	0.46	0.49	0.44	0.21
Sisben	0.62	0.56	0.63	0.01	0.65
<i>Panel B: Education</i>					
Primary or less	0.19	0.07	0.12	0.00	0.01
Secondary	0.49	0.61	0.49	0.00	1.00
Tertiary	0.32	0.32	0.39	0.34	0.06
<i>Panel C: Labor market</i>					
Employed at baseline	0.33	0.52	0.35	0.00	0.56
Wage (USD)	46.54	77.53	54.70	0.00	0.25
Log wage	4.17	6.18	4.53	0.00	0.51
Reservation Wage (USD)	367.79	356.96	357.41	0.06	0.30
<i>Panel D: Experience</i>					
Less than 1 year	0.13	0.10	0.09	0.10	0.01
1-2 years	0.18	0.27	0.18	0.01	0.76
3-5 years	0.20	0.28	0.20	0.01	0.87
6-10 years	0.16	0.16	0.18	0.59	0.53
<i>Panel E: Competency level</i>					
Recycling	0.07	0.06	0.06	0.60	0.80
Health Services	0.45	0.14	0.41	0.00	0.31
Food Manipulation	0.27	0.52	0.29	0.00	0.43
Design and tailoring	0.09	0.08	0.09	0.65	0.94
Food Preparation	0.12	0.18	0.14	0.00	0.37
Observations	1,676	484	484		

Notes: This table reports baseline characteristics for locals and migrants. Column (3) reports weighted means for migrants after re-weighting using normalized inverse propensity-score weights (estimated by logit on baseline covariates) so that the distribution of observable characteristics among migrants matches that of locals. Columns (4) and (5) report p-values from regressions of each covariate on an indicator for migrant status, estimated unweighted and with reweighting, respectively. In the reweighted specification, locals retain a weight of one and only the migrant sub-sample is reweighted by the normalized inverse propensity-score weight. All regressions include competency fixed effects and use robust standard errors, except for the competency share variables in Panel E, where the fixed effects coincide with the outcome and are therefore omitted.

Appendix Table G.3.2: Effects of Access to Skills Certification on Labor Market Outcomes

Migrants Reweighted to Local Baseline Composition
 Intention To Treat Effects Estimated using Equation (1)

Outcomes Replaced by Zero if Non-Employed							
	Employed	Inactive	Log Wage	Formal emp.	Weekly working hours	Full time (7+ hrs)	Skill Downgrade
<i>Panel A: Full Sample</i>							
Treatment	0.037* (0.021)	-0.027 (0.017)	0.476 (0.294)	-0.012 (0.016)	0.652 (0.931)	0.025 (0.020)	-0.015 (0.019)
Observations	5,647	5,647	5,634	5,647	5,647	5,647	5,647
Mean Control	0.521	0.248	13.42	0.258	28.17	0.481	0.319
<i>Panel B: Locals</i>							
Treatment	0.057*** (0.021)	-0.035* (0.018)	0.741*** (0.283)	-0.008 (0.015)	2.016** (0.857)	0.037** (0.017)	0.004 (0.016)
Observations	4,404	4,404	4,392	4,404	4,404	4,404	4,404
Mean Control	0.494	0.264	13.34	0.279	26.48	0.448	0.281
<i>Panel C: Migrants</i>							
Treatment	-0.023 (0.062)	-0.005 (0.047)	-0.332 (0.869)	-0.018 (0.041)	-3.832 (2.724)	-0.007 (0.066)	-0.067 (0.066)
Observations	1,243	1,243	1,242	1,243	1,243	1,243	1,243
Mean Control	0.646	0.177	13.68	0.188	33.99	0.592	0.449
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. The table reports pooled ITT estimates from Equation 1, estimated by weighted least squares using normalized inverse-odds weights that align the baseline covariate distribution of migrants to that of locals. Weights equal $(1 - \hat{p})/\hat{p}$ for migrants and one for locals, where $\hat{p}$ is the predicted probability of being a migrant from a baseline logit on age and its square, sex, household headship, dependent children, caregiving status, education, Sisben status, prior employment, prior wage and log wage, work experience, and competence; migrant weights are rescaled to unit mean so that the weighted and unweighted migrant sample sizes coincide. Panel A presents results for the full sample; Panels B and C report estimates separately for locals and migrants. Outcomes in columns (3)–(7) are replaced by zero if the outcome is missing for non-employed individuals. For non-employed individuals, monthly earnings are set to zero prior to the transformation $\text{Log}(\text{Wage} + 1)$. All specifications include competence and wave fixed effects; full-sample regressions additionally include a migrant indicator. Standard errors (in parentheses) are clustered at the individual level. Mean Control reports the weighted control-group mean in the corresponding estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

H Secondary Outcomes

Appendix Table H.1: Effects of Access to Skill Certification on Stated Preferences

Intention To Treat Effects Estimated using Equation (1)

	Participate w/o subsidy	Willing to pay	WTP (USD, midpoint)
<i>Panel A: Full Sample</i>			
Treatment	0.004 (0.015)	-0.016 (0.019)	0.314 (0.744)
Observations	1,944	3,344	3,344
Mean Control	0.882	0.637	19.10
<i>Panel B: Locals</i>			
Treatment	-0.009 (0.018)	-0.009 (0.023)	0.707 (0.828)
Observations	1,512	2,569	2,569
Mean Control	0.880	0.614	18.65
<i>Panel C: Migrants</i>			
Treatment	0.045 (0.028)	-0.032 (0.036)	-0.318 (1.611)
Observations	432	775	775
Mean Control	0.890	0.740	21.09
Competence FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes

Notes: Treatment is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. The table reports pooled ITT estimates from Equation 1. Panel A presents results for the full sample; Panels B and C report estimates separately for locals and migrants. Willingness to participate without subsidies was collected at Follow-up 1. Willingness to pay was collected at Follow-ups 1 and 3. All specifications include competence and wave fixed effects and control for gender, age, age squared, and education; full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Mean Control reports the control-group mean in the estimation sample. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix Table H.2: Effects of Access to Skills Certification on Firm Characteristics and Job Mobility

Intention To Treat Effects Estimated using Equation (1)

	Solo/ self-emp.	Small firm	Med./large firm	Changed sector	Changed occupation	Changed job	Number of job offers
<i>Panel A: Full Sample</i>							
Treatment	0.025* (0.014)	0.017 (0.016)	-0.005 (0.010)	0.005 (0.011)	0.012 (0.012)	0.019 (0.014)	-0.051** (0.024)
Observations	5,647	5,647	5,647	5,647	5,647	5,647	5,549
Mean Control	0.177	0.254	0.0835	0.0895	0.119	0.161	0.221
<i>Panel B: Locals</i>							
Treatment	0.041** (0.016)	0.032* (0.017)	-0.013 (0.011)	0.021* (0.012)	0.016 (0.013)	0.030** (0.015)	-0.039 (0.026)
Observations	4,404	4,404	4,404	4,404	4,404	4,404	4,333
Mean Control	0.169	0.228	0.0814	0.0803	0.106	0.136	0.200
<i>Panel C: Migrants</i>							
Treatment	-0.029 (0.030)	-0.029 (0.037)	0.020 (0.023)	-0.046* (0.024)	-0.003 (0.031)	-0.020 (0.033)	-0.094* (0.056)
Observations	1,243	1,243	1,243	1,243	1,243	1,243	1,216
Mean Control	0.219	0.377	0.0938	0.134	0.182	0.286	0.323
Competence FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Assigned to certification (ITT) is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. The table reports pooled ITT estimates from Equation 1. Panel A presents results for the full sample. Panels B and C report estimates separately for locals and migrants. Firm size outcomes (columns 1–3) are set to zero for respondents not employed at the follow-up. Sector change (column 4), occupation change (column 5), and job change (column 6) are set to zero for respondents not employed at baseline or follow-up. Number of job offers is set to zero for respondents reporting no offers. All specifications include competence and wave fixed effects and control for gender, age, age squared, and education. Full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Mean Control reports the control-group mean in the estimation sample. *** p<0.01, ** p<0.05, * p<0.1.

Appendix Table H.3: Effects of Access to Skills Certification on Job Satisfaction and Well-Being

Intention To Treat Effects Estimated using Equation (1)

	Job satisfaction	Skipping a meal	Anxiety (MS)	Anxiety (GAD-7)
<i>Panel A: Full Sample</i>				
Treatment	0.019 (0.018)	0.011 (0.018)	-0.014 (0.021)	-0.110 (0.246)
Observations	3,667	5,549	1,980	1,944
Mean Control	0.290	0.302	0.301	6.548
<i>Panel B: Locals</i>				
Treatment	0.030 (0.020)	0.025 (0.021)	-0.031 (0.024)	-0.143 (0.292)
Observations	2,866	4,333	1,538	1,512
Mean Control	0.268	0.304	0.329	6.940
<i>Panel C: Migrants</i>				
Treatment	-0.017 (0.041)	-0.034 (0.035)	0.040 (0.037)	0.010 (0.448)
Observations	801	1,216	442	432
Mean Control	0.395	0.293	0.167	4.680
Competence FE	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes

Notes: Assigned to certification (ITT) is an indicator equal to one for individuals randomly assigned to receive a certification offer, regardless of completion. The table reports pooled ITT estimates from Equation 1. Panel A presents results for the full sample; Panels B and C report estimates separately for locals and migrants. Job satisfaction is a binary indicator (satisfied or very satisfied) collected among employed respondents at Follow-ups 2–3. Anxiety (MS) and Anxiety (GAD-7) are collected at Follow-up 1 only. All specifications include competence and wave fixed effects and control for gender, age, age squared, and education. Full-sample regressions additionally control for migrant status. Standard errors (in parentheses) are clustered at the individual level. Mean Control reports the control-group mean in the estimation sample. *** p<0.01, ** p<0.05, * p<0.1.